Agentic Software Engineering System

URL Shortener + DAG-orchestrated agent platform: web console walkthrough and API execution report

Generated	2026-11-08 09:42:40 UTC
Application	Spring Boot 3.4.1, Java 21 target, H2 (file) database, HTTP Basic security
LLM	MockLLMProvider (deterministic simulation, no external API)
Screens captured	46 screenshots, one per step, driven through the real web console
API calls in this report	328 calls made by the browser
API endpoints covered	29 of 29 endpoints in the OpenAPI specification
Run-all-flows result	**72 passed, 0 failed** (72 checks)
Automated test suite	**91 tests, 0 failures** (mvn verify, last run in the project)

All content generated by the platform's agents is a deterministic simulation produced by the mock LLM; it is not real model output.

Contents

1. **Sign-in and dashboard** (steps 1-2)
2. **URL shortener** (steps 3-8)
3. **Scenario 1 - Greenfield** (steps 9-17)
4. **Scenario 2 - Brownfield** (steps 18-21)
5. **Scenario 3 - Ambiguous requirement** (steps 22-26)
6. **Rejection and rollback** (steps 27-28)
7. **Safe-stop (security)** (steps 29-32)
8. **Retry, fallback and fault injection** (steps 33-36)
9. **Dynamic replanning** (steps 37-41)
10. **Observability** (steps 42-42)
11. **Run all flows from the browser** (steps 43-46)
12. **Appendix A** - Complete list of checks executed by "Run all flows"
13. **Appendix B** - API coverage against the OpenAPI specification

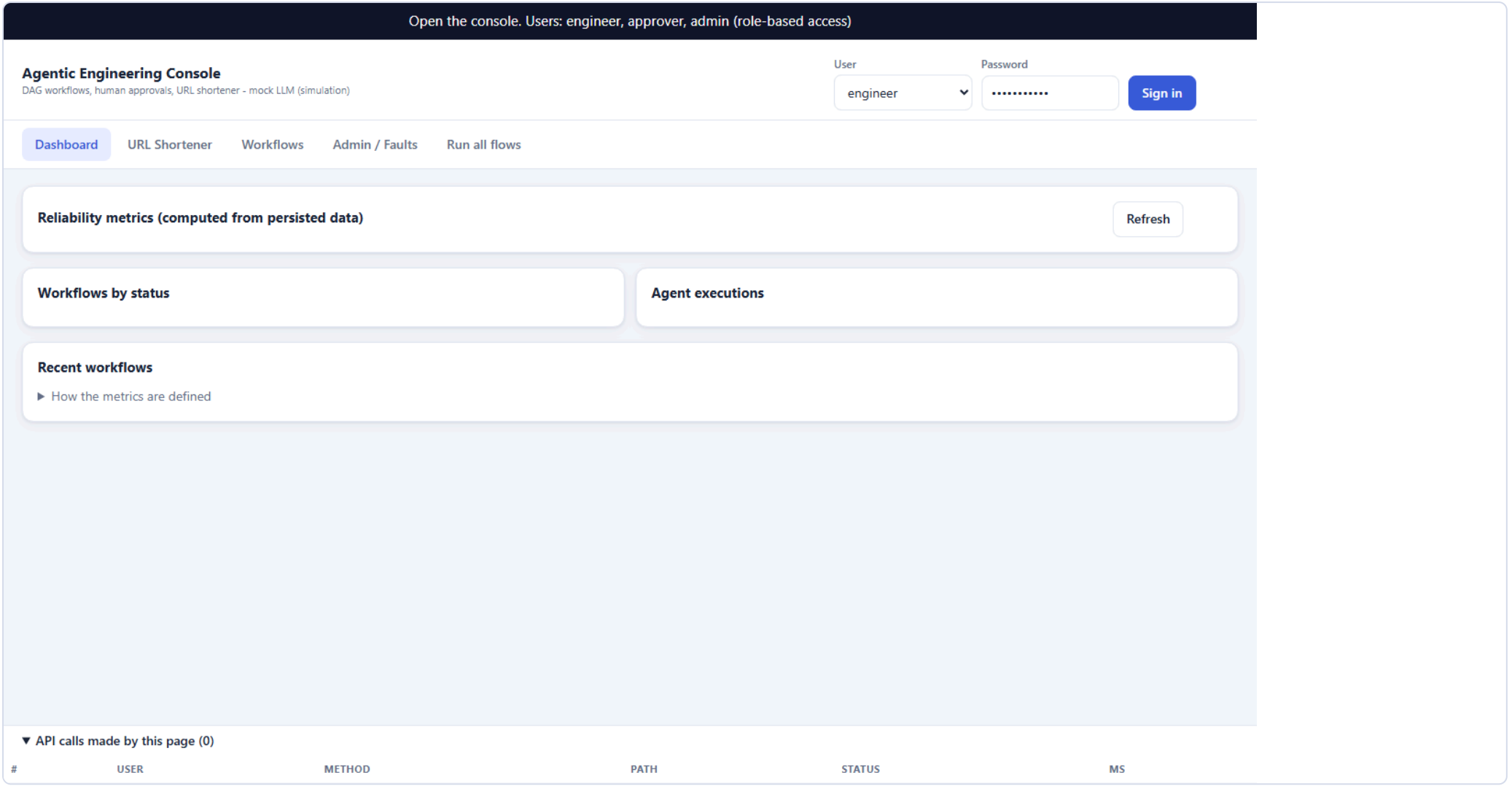
How the flows were executed

A headless browser drove the console exactly as a user would: signing in as *engineer*, *approver* and *admin*, creating URLs and workflows, executing them, approving or rejecting at the human gates, injecting faults, replanning and reading metrics. Each step below shows the screen and the REST calls the page made (user, method, path, HTTP status, latency). The final steps run every API flow from the browser in one go.

SIGN-IN AND DASHBOARD

Step 1. Console before sign-in

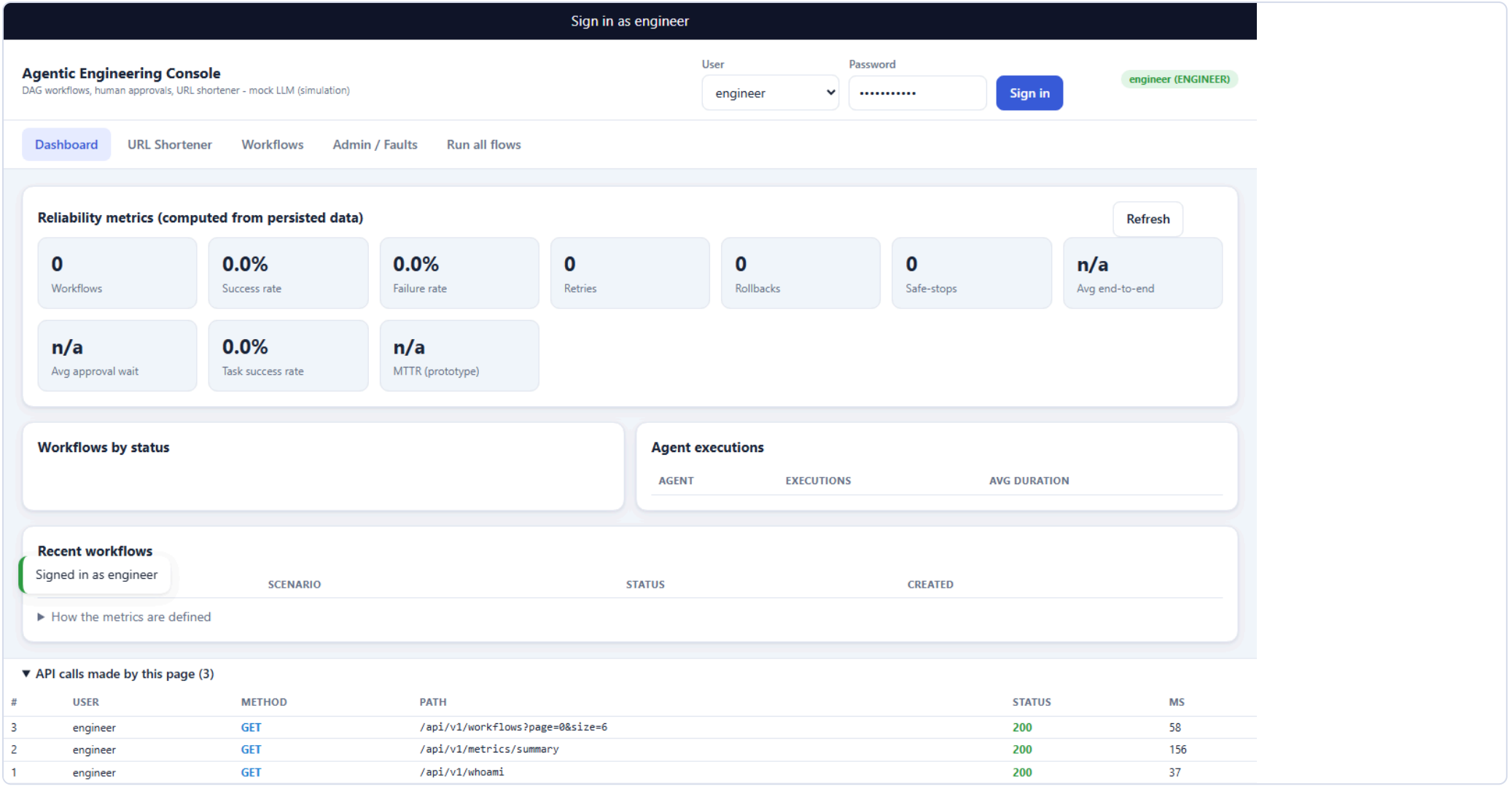
The single-page console is served by the Spring Boot application itself (no separate frontend server). Every API call the page makes is listed in the panel at the bottom.



SIGN-IN AND DASHBOARD

Step 2. Signed in as engineer, dashboard

GET /api/v1/whoami identifies the user and roles; the dashboard reads GET /api/v1/metrics/summary. The database is fresh, so all counters start at zero.



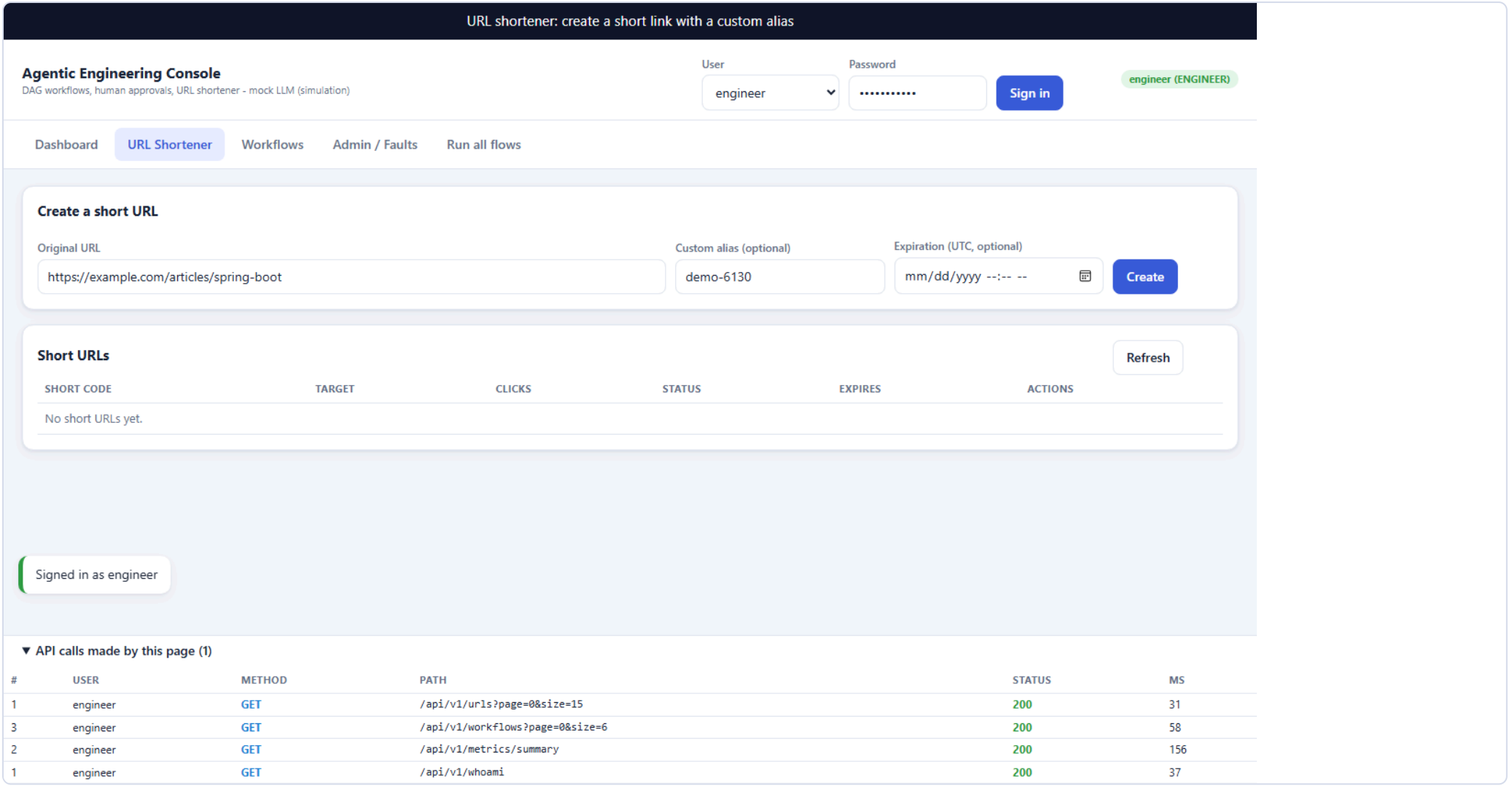
USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/whoami	200	37
engineer	GET	/api/v1/metrics/summary	200	156

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows?page=0&size=6	200	58

URL SHORTENER

Step 3. Create form

Original URL, optional custom alias and optional expiration date.

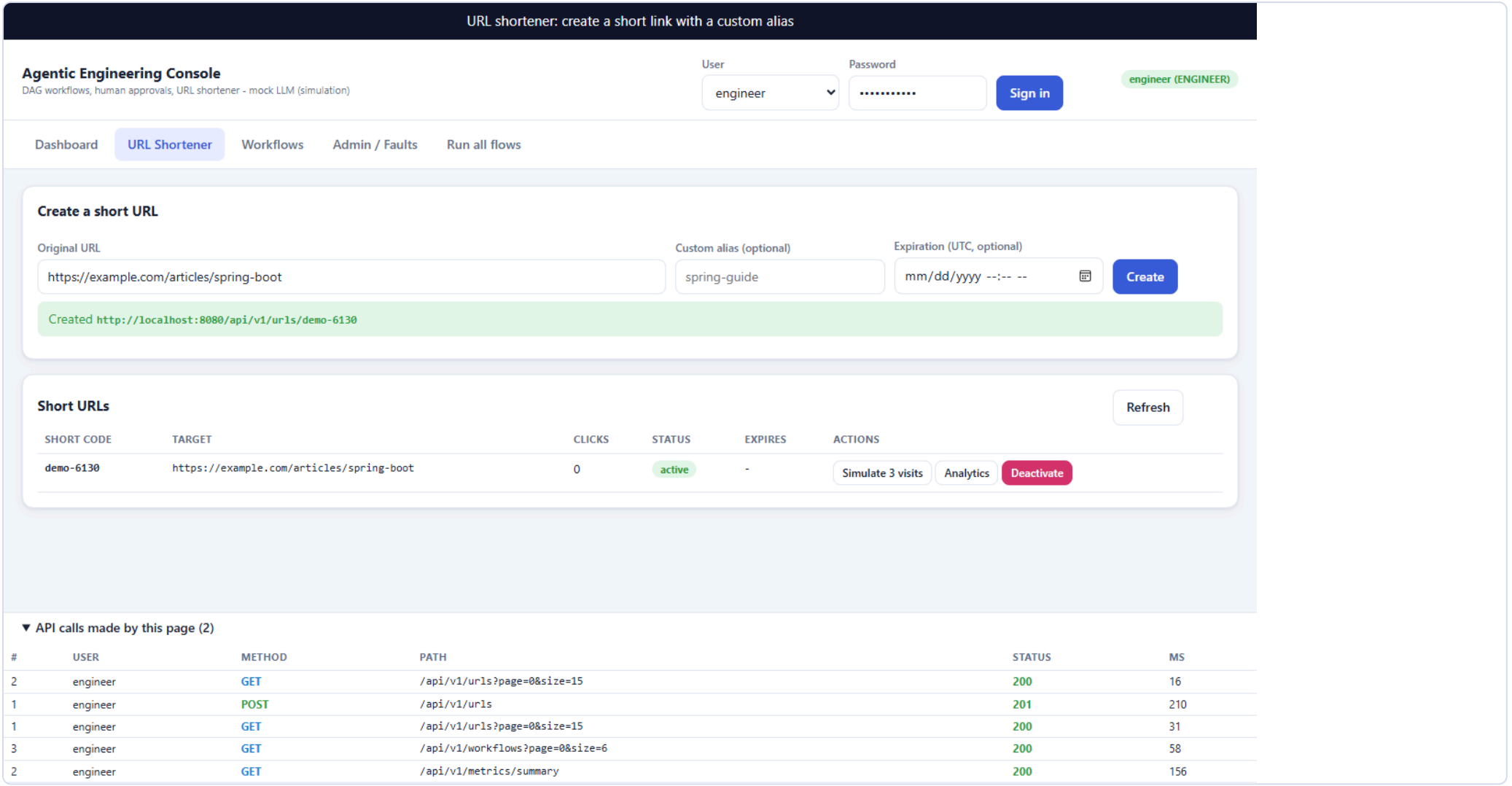


USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/urls?page=0&size=15	200	31

URL SHORTENER

Step 4. Short URL created

POST /api/v1/urls returned 201; GET /api/v1/urls refreshed the table.

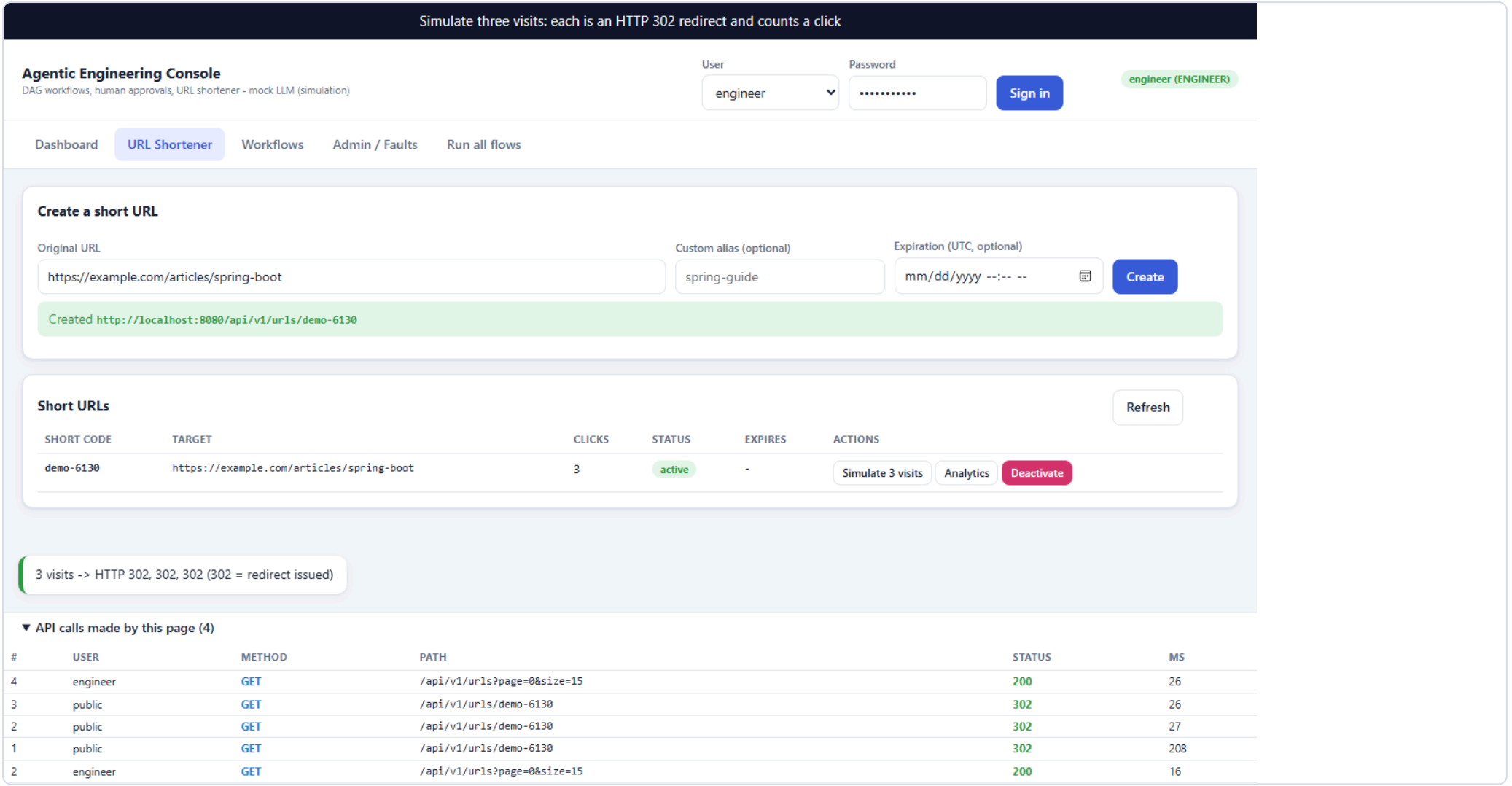


USER	METHOD	API CALL	HTTP	MS
engineer	POST	/api/v1/urls	201	210
engineer	GET	/api/v1/urls?page=0&size=15	200	16

URL SHORTENER

Step 5. Three redirects recorded

GET /api/v1/urls/{code} answers 302 (the browser reports it as an opaque redirect). Click count and timestamps are persisted.



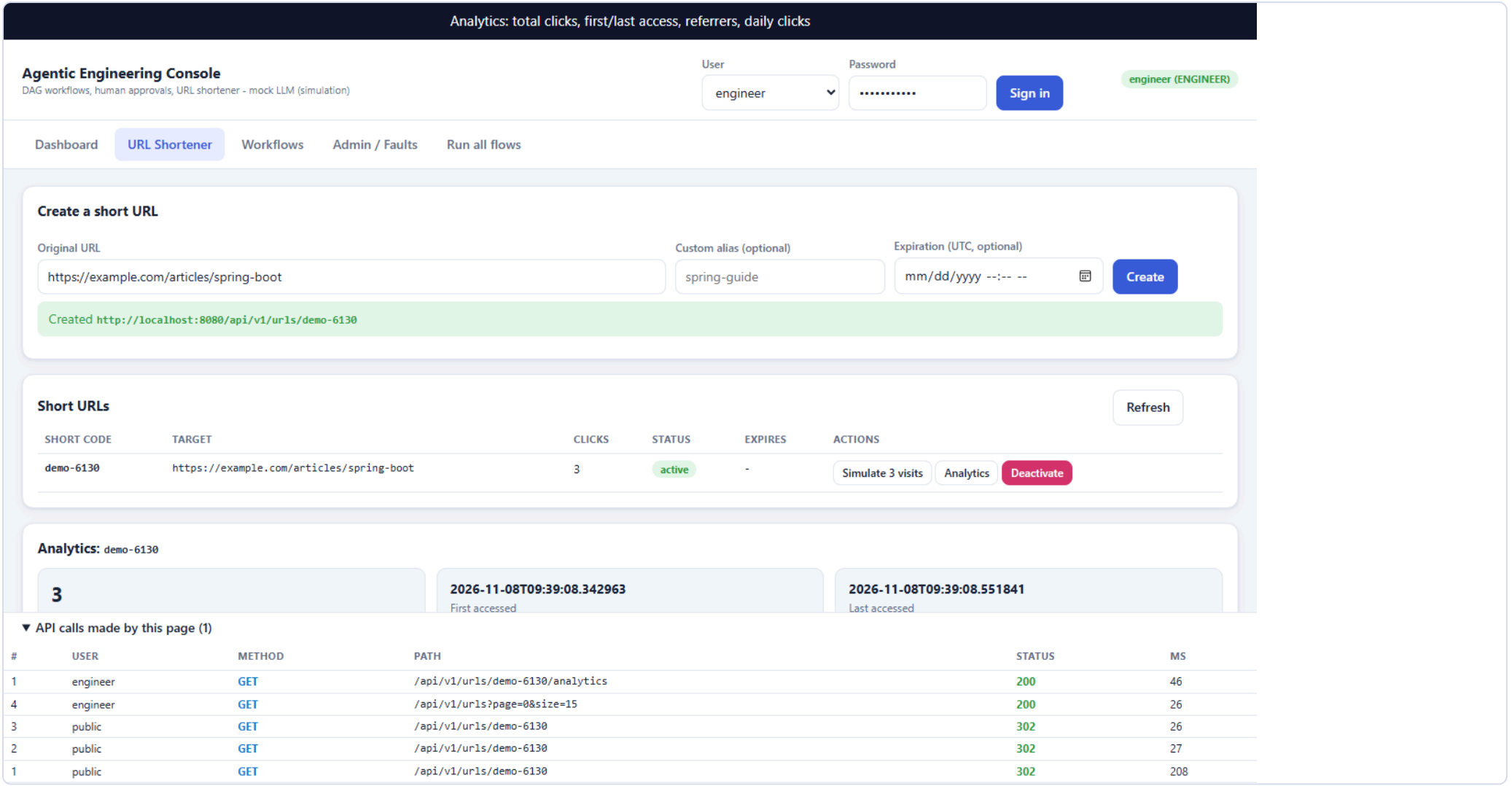
USER	METHOD	API CALL	HTTP	MS
public	GET	/api/v1/urls/demo-6130	302	208
public	GET	/api/v1/urls/demo-6130	302	27

USER	METHOD	API CALL	HTTP	MS
public	GET	/api/v1/urls/demo-6130	302	26
engineer	GET	/api/v1/urls?page=0&size=15	200	26

URL SHORTENER

Step 6. Analytics

GET /api/v1/urls/{code}/analytics: total clicks, first and last access timestamps, referrer statistics and daily click statistics.

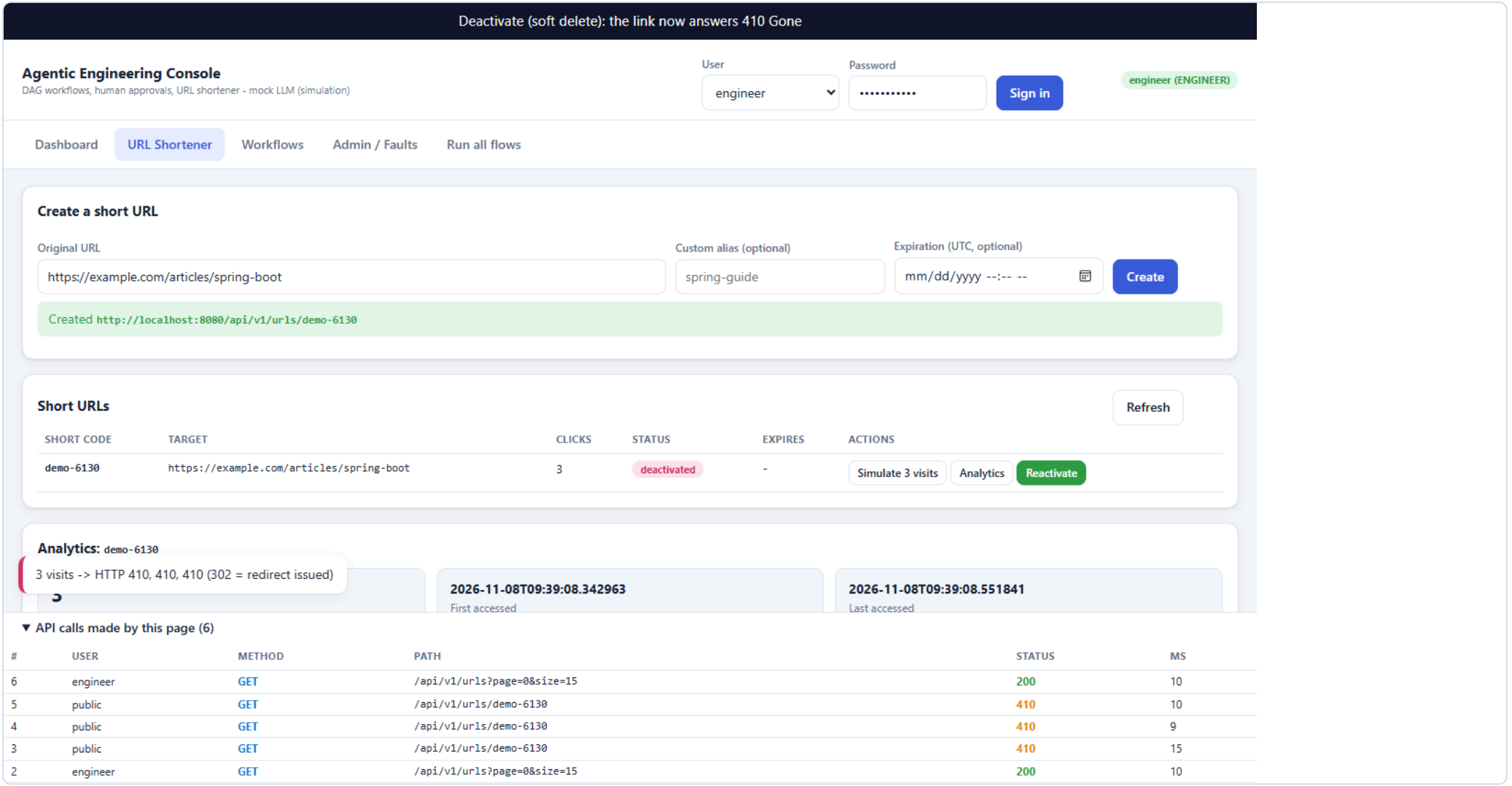


USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/urls/demo-6130/analytics	200	46

URL SHORTENER

Step 7. Deactivated link answers 410

DELETE /api/v1/urls/{code} returned 204 (soft delete); visiting the link now returns 410 Gone.



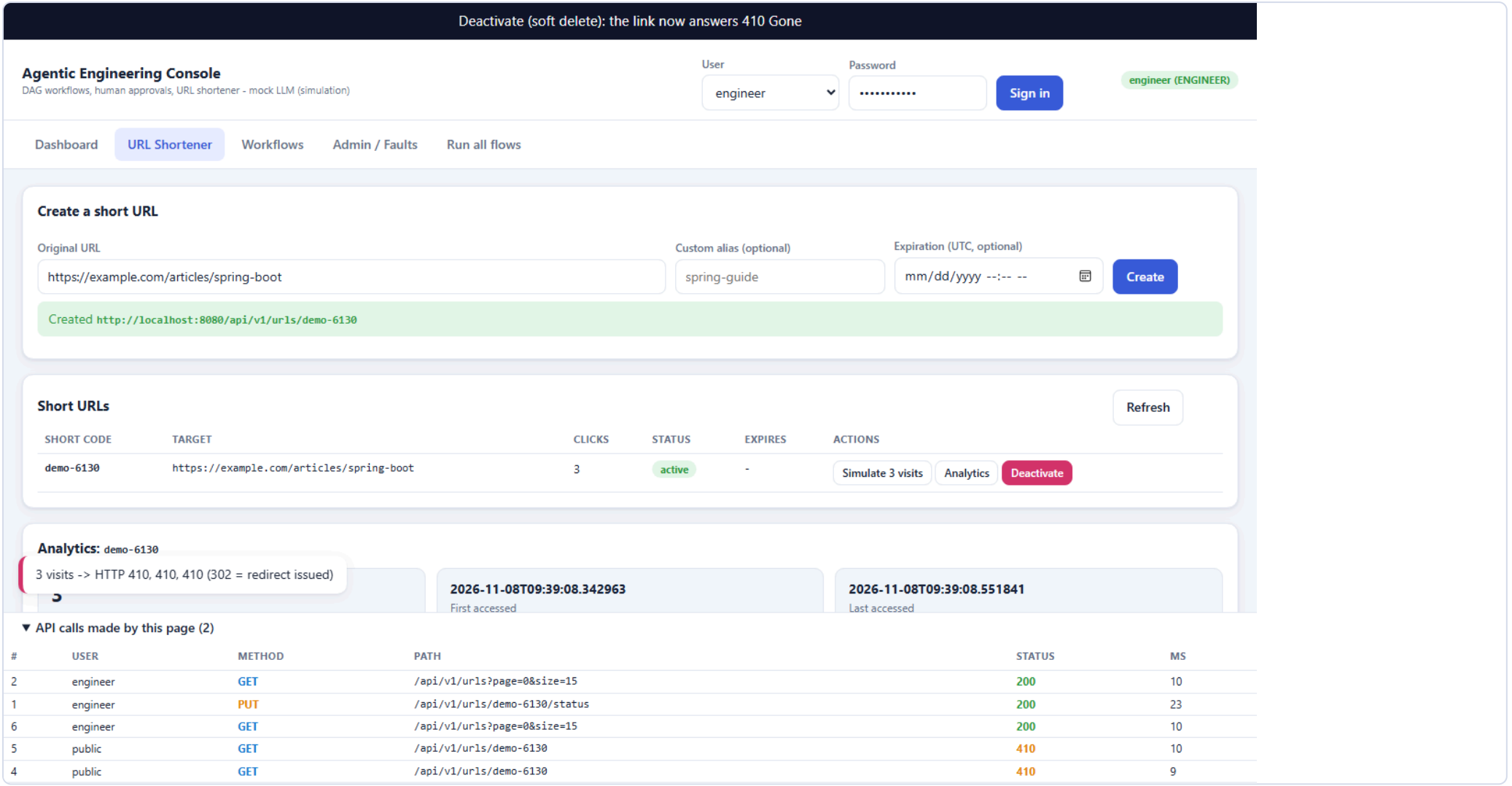
USER	METHOD	API CALL	HTTP	MS
engineer	DELETE	/api/v1/urls/demo-6130	204	20
engineer	GET	/api/v1/urls?page=0&size=15	200	10

USER	METHOD	API CALL	HTTP	MS
public	GET	/api/v1/urls/demo-6130	410	15
public	GET	/api/v1/urls/demo-6130	410	9
public	GET	/api/v1/urls/demo-6130	410	10
engineer	GET	/api/v1/urls?page=0&size=15	200	10

URL SHORTENER

Step 8. Link reactivated

PUT /api/v1/urls/{code}/status {active:true} reactivated the link.

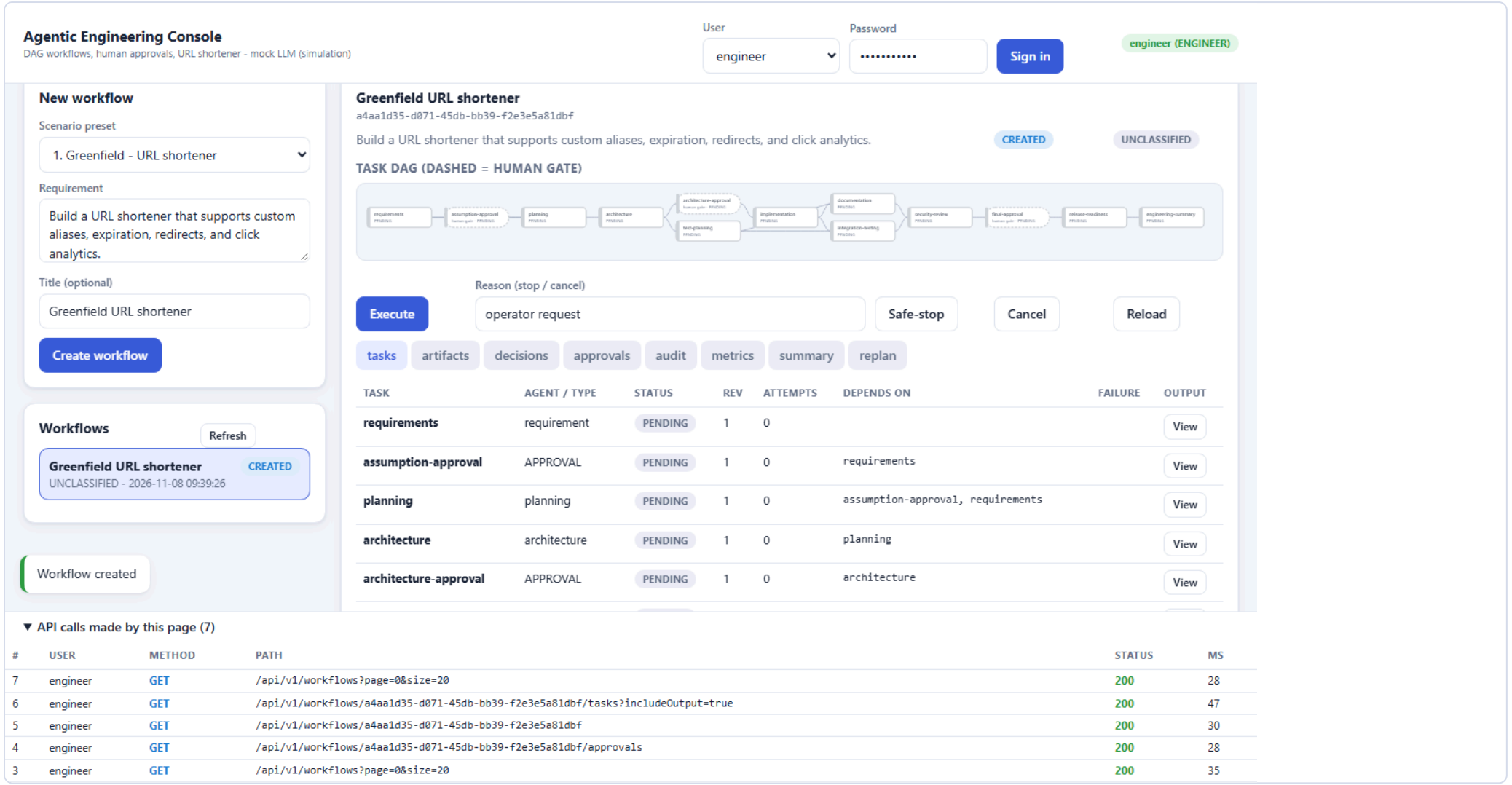


USER	METHOD	API CALL	HTTP	MS
engineer	PUT	/api/v1/urls/demo-6130/status	200	23
engineer	GET	/api/v1/urls?page=0&size=15	200	10

SCENARIO 1 - GREENFIELD

Step 9. Workflow created

POST /api/v1/workflows created the workflow with 13 DAG nodes (nothing has run yet). Dashed nodes are human approval gates.



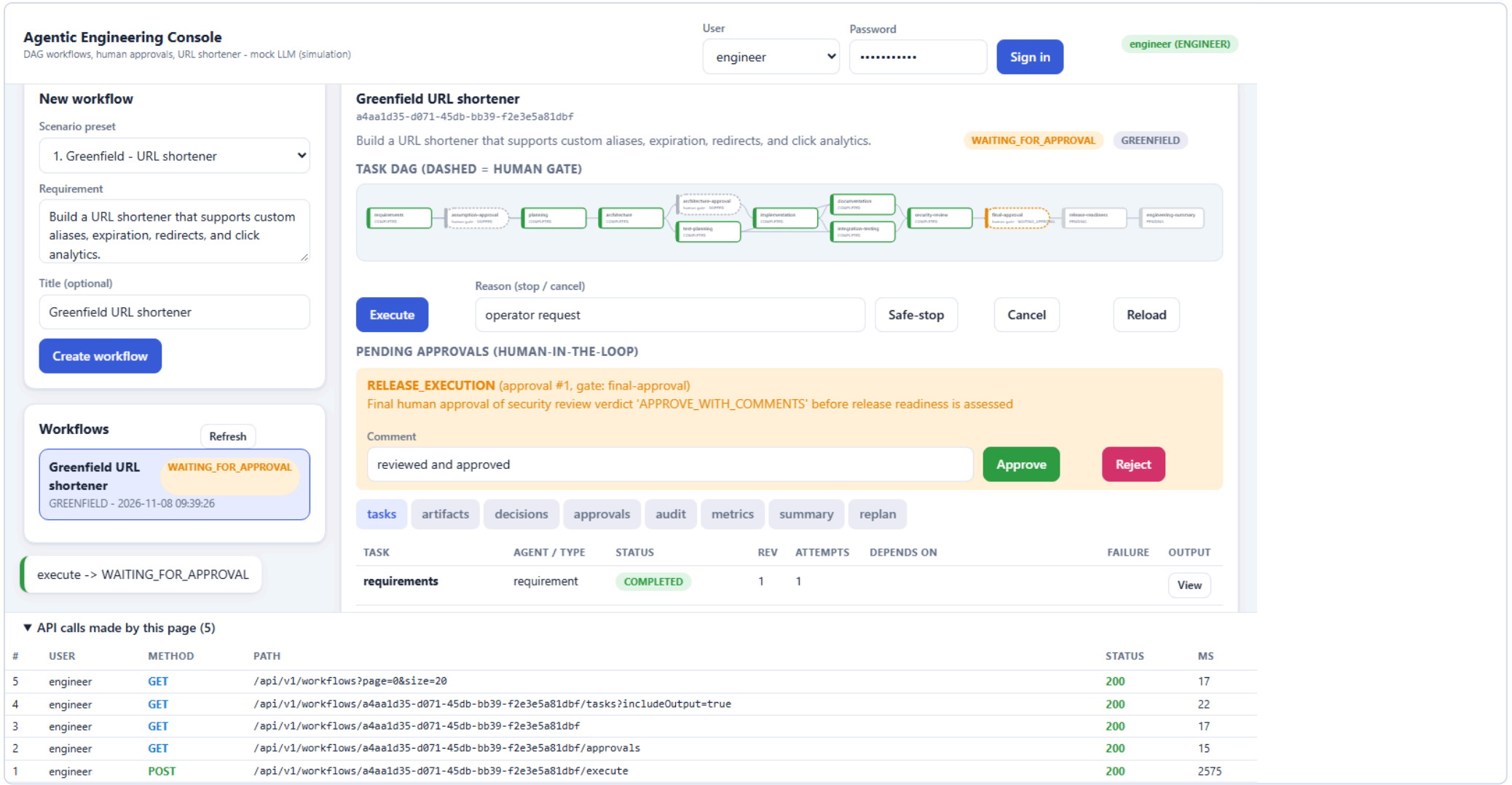
USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows?page=0&size=20	200	21
engineer	POST	/api/v1/workflows	201	471

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows?page=0&size=20	200	35
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approvals	200	28
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	30
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/tasks?includeOutput=true	200	47
engineer	GET	/api/v1/workflows?page=0&size=20	200	28

SCENARIO 1 - GREENFIELD

Step 10. Executed, waiting for final approval

POST /execute ran requirement, planning, architecture, then implementation and test-planning in parallel, documentation and validation, security review. The workflow paused at the human final approval; the two conditional gates were skipped.



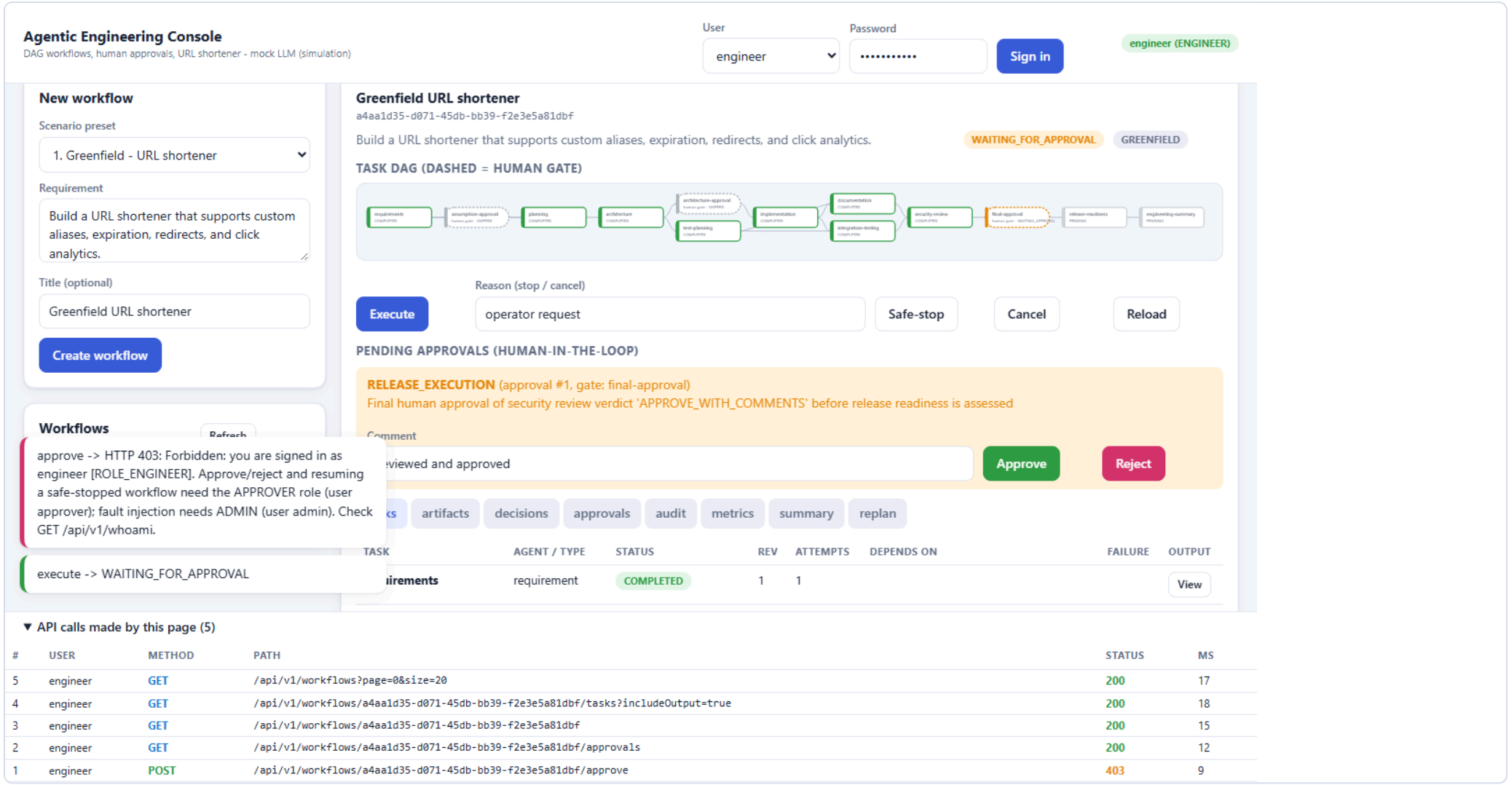
USER	METHOD	API CALL	HTTP	MS
engineer	POST	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/execute	200	2575
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approvals	200	15

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	17
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/tasks?includeOutput=true	200	22
engineer	GET	/api/v1/workflows?page=0&size=20	200	17

SCENARIO 1 - GREENFIELD

Step 11. Engineer approval denied (403)

The server answered 403 and explains which role is needed. Approvals are enforced by role (APPROVER).



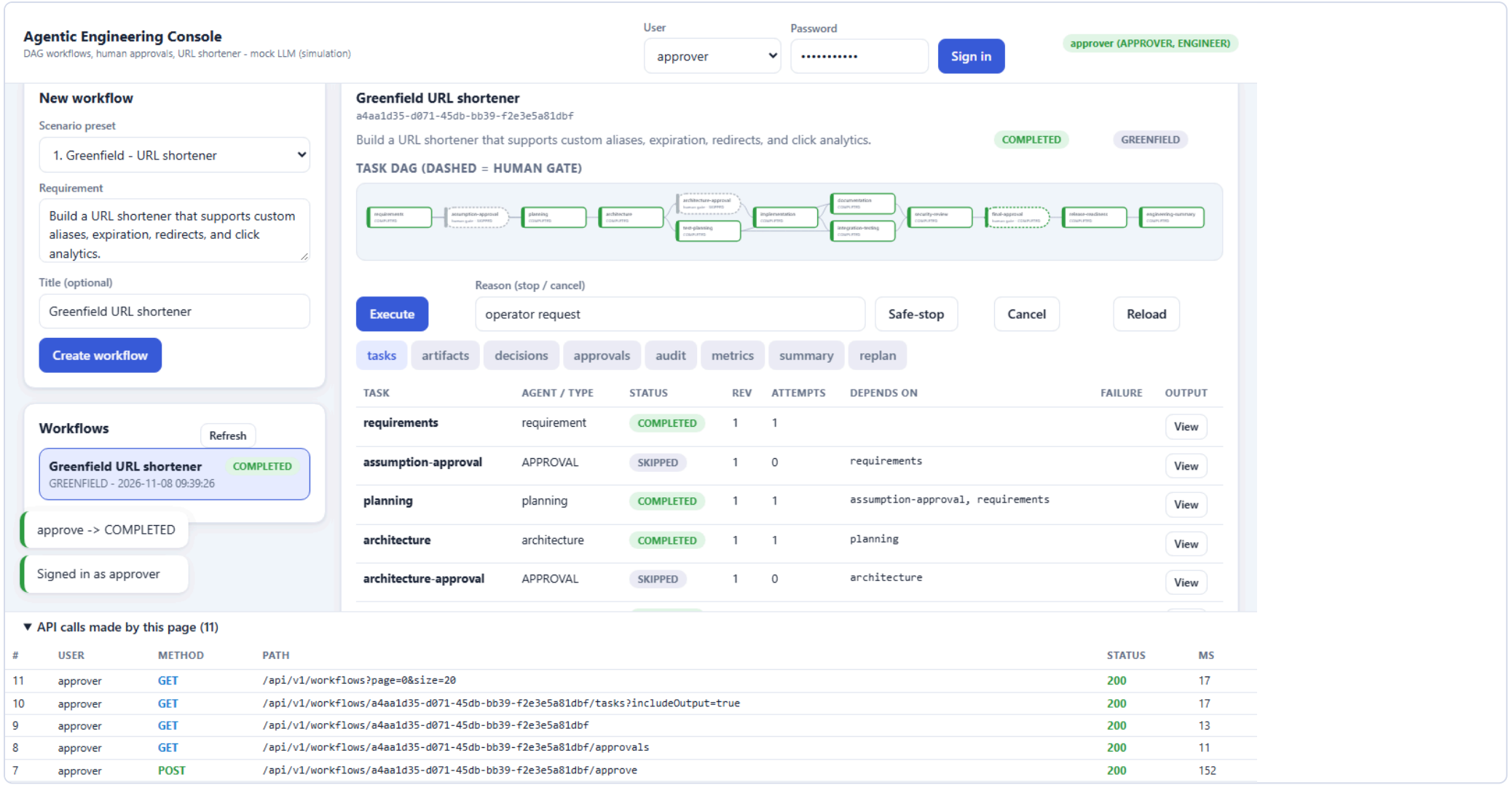
USER	METHOD	API CALL	HTTP	MS
engineer	POST	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approve	403	9
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approvals	200	12

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	15
engineer	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/tasks?includeOutput=true	200	18
engineer	GET	/api/v1/workflows?page=0&size=20	200	17

SCENARIO 1 - GREENFIELD

Step 12. Approved, workflow completed

POST /approve by the approver granted the gate; the workflow resumed and COMPLETED (release readiness and engineering summary).



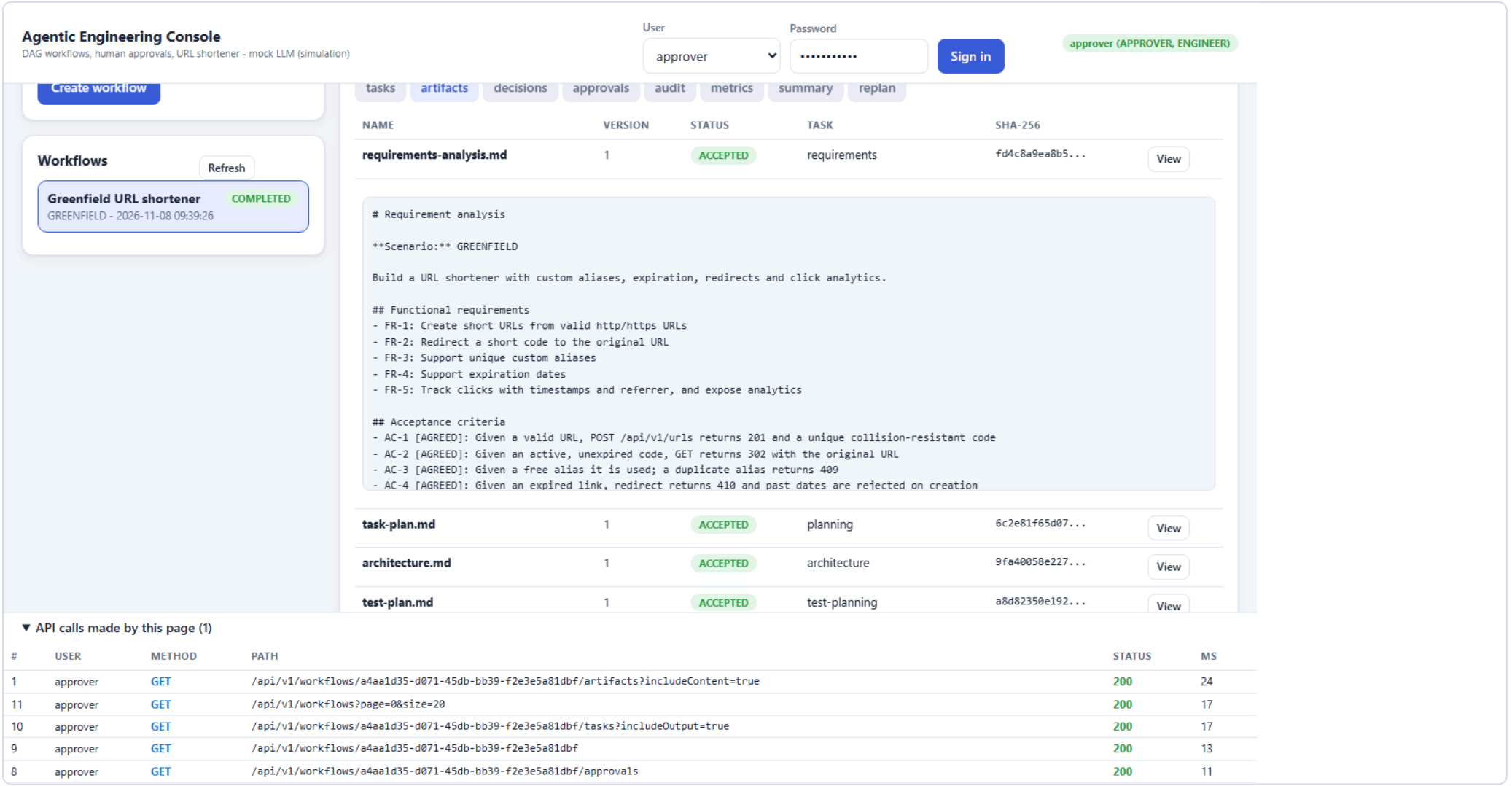
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/whoami	200	10
approver	GET	/api/v1/workflows?page=0&size=20	200	22

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approvals	200	9
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	14
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/tasks?includeOutput=true	200	15
approver	GET	/api/v1/workflows?page=0&size=20	200	17
approver	POST	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approve	200	152
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/approvals	200	11
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	13
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/tasks?includeOutput=true	200	17
approver	GET	/api/v1/workflows?page=0&size=20	200	17

SCENARIO 1 - GREENFIELD

Step 13. Generated artifacts

GET /artifacts?includeContent=true: versioned artifacts, all ACCEPTED after final approval; content stored in the approved workspace only.

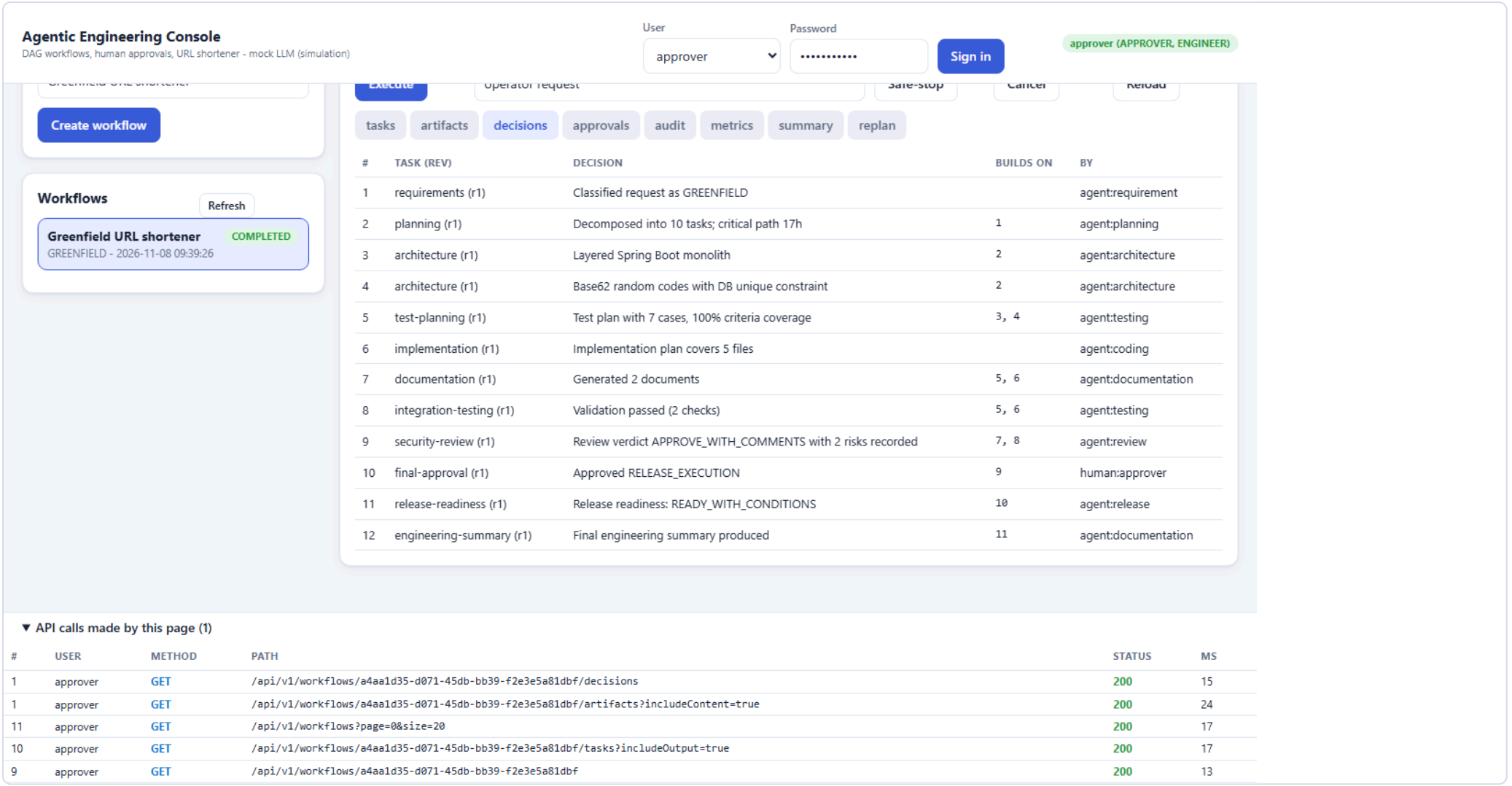


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/artifacts?includeContent=true	200	24

SCENARIO 1 - GREENFIELD

Step 14. Decision lineage

GET /decisions: each decision lists the upstream decisions it builds on (Builds on column).

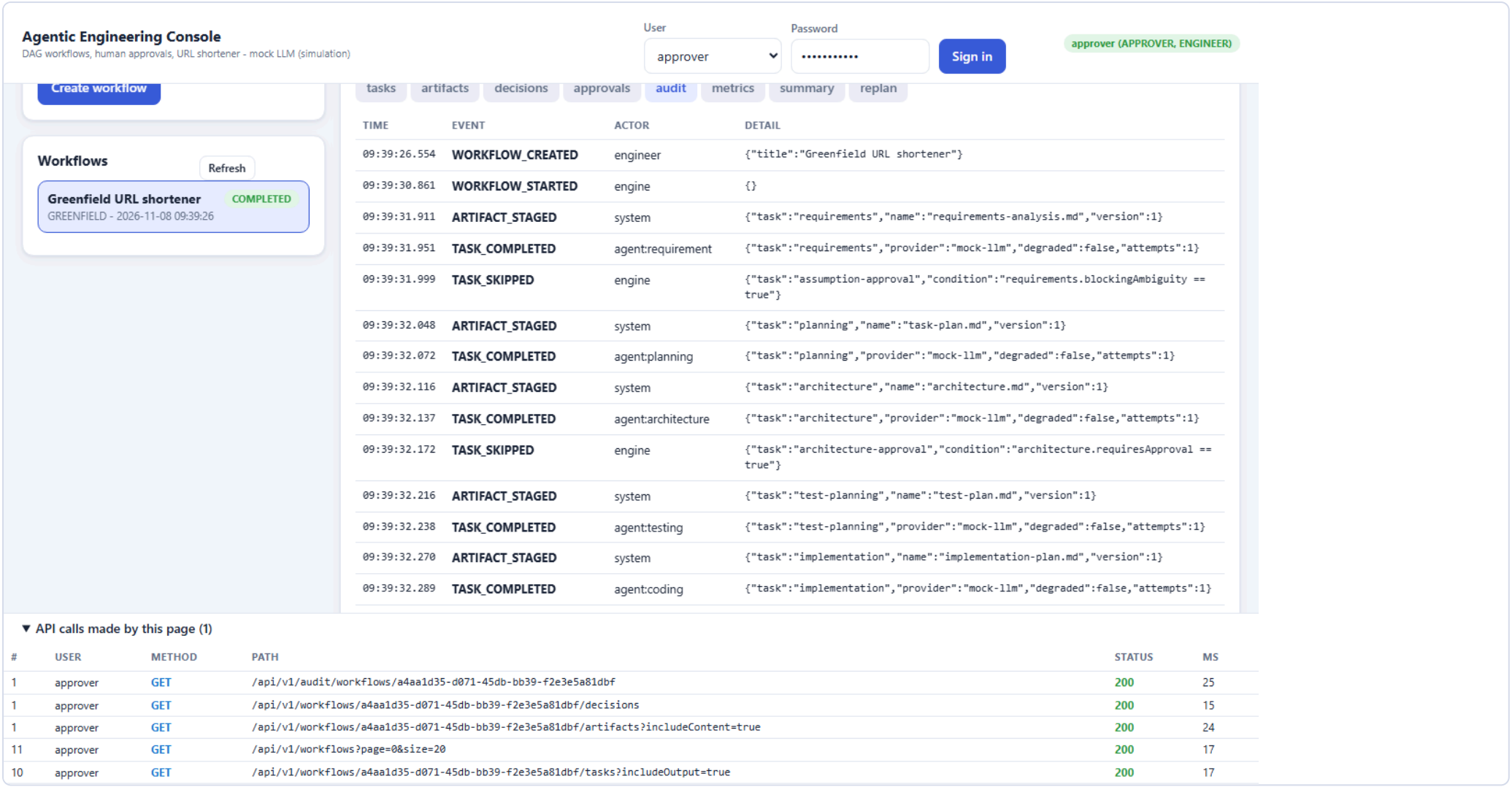


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/decisions	200	15

SCENARIO 1 - GREENFIELD

Step 15. Audit trail

GET /api/v1/audit/workflows/{id}: every sensitive step is recorded (creation, task results, approvals, artifact promotion).

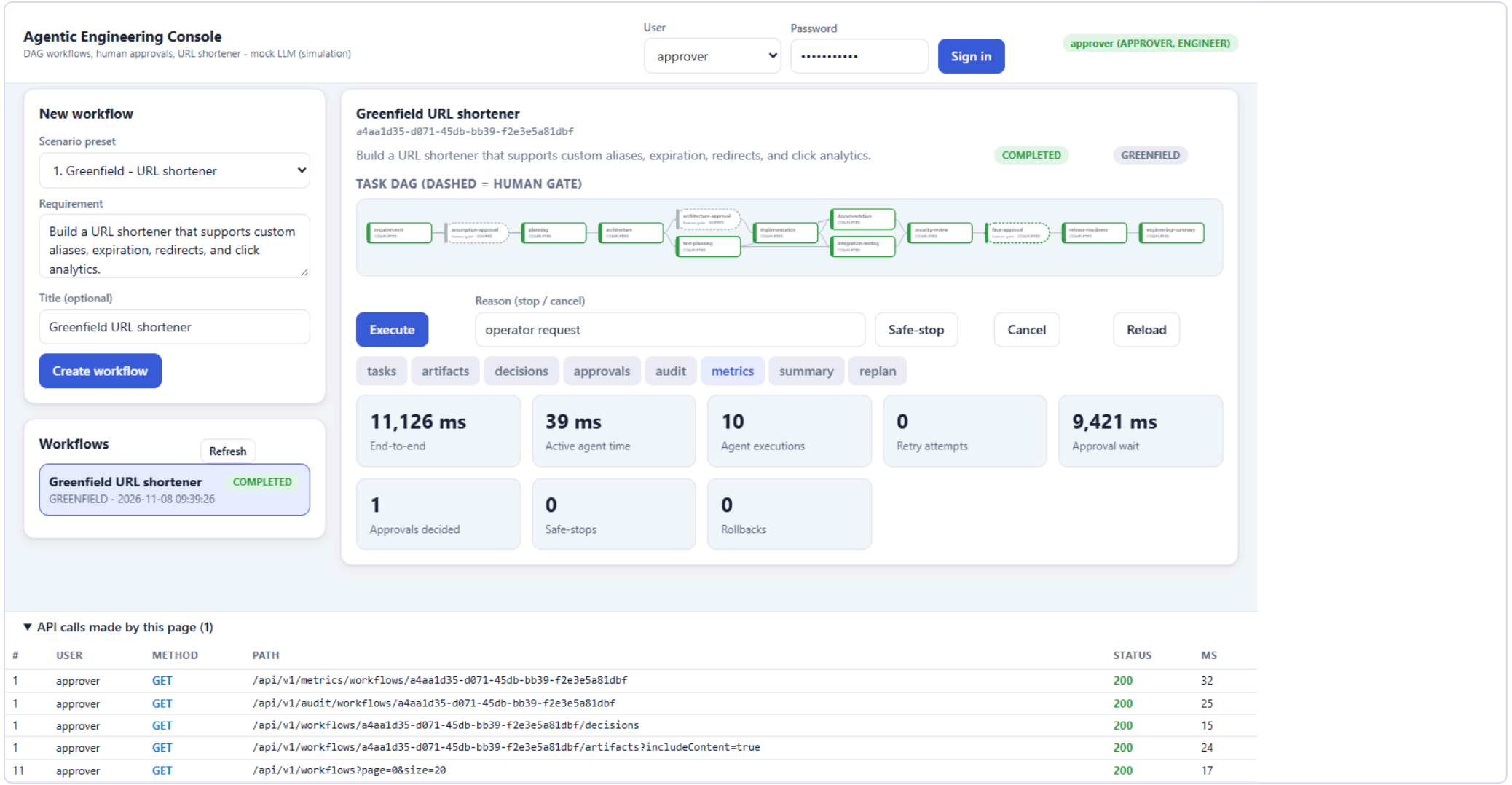


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/audit/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	25

SCENARIO 1 - GREENFIELD

Step 16. Workflow metrics

GET /api/v1/metrics/workflows/{id}: latency, agent executions, retries, approval wait.

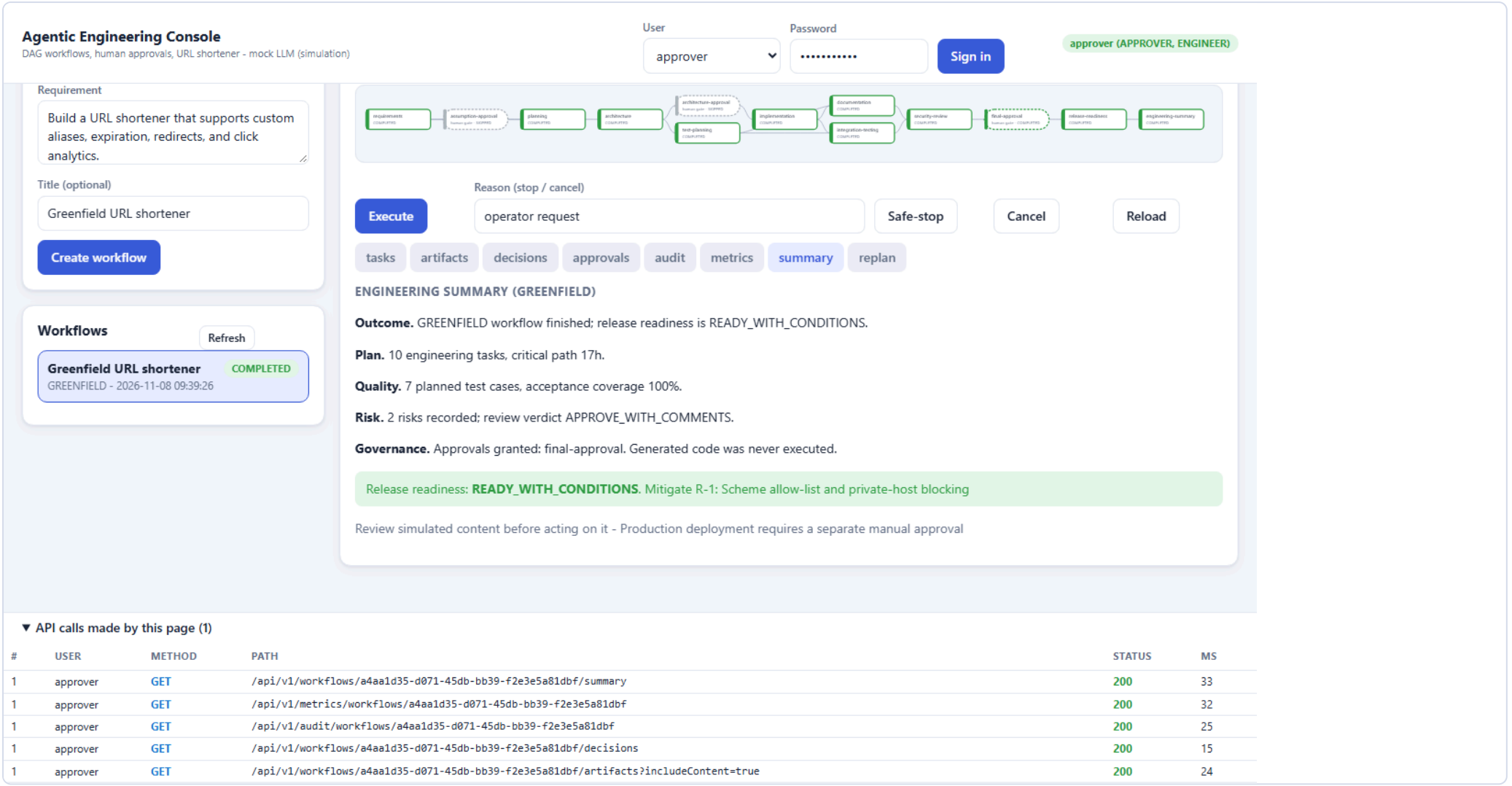


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/metrics/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf	200	32

SCENARIO 1 - GREENFIELD

Step 17. Engineering summary

GET /summary: final engineering summary and release readiness. The platform never deploys; it only assesses readiness.

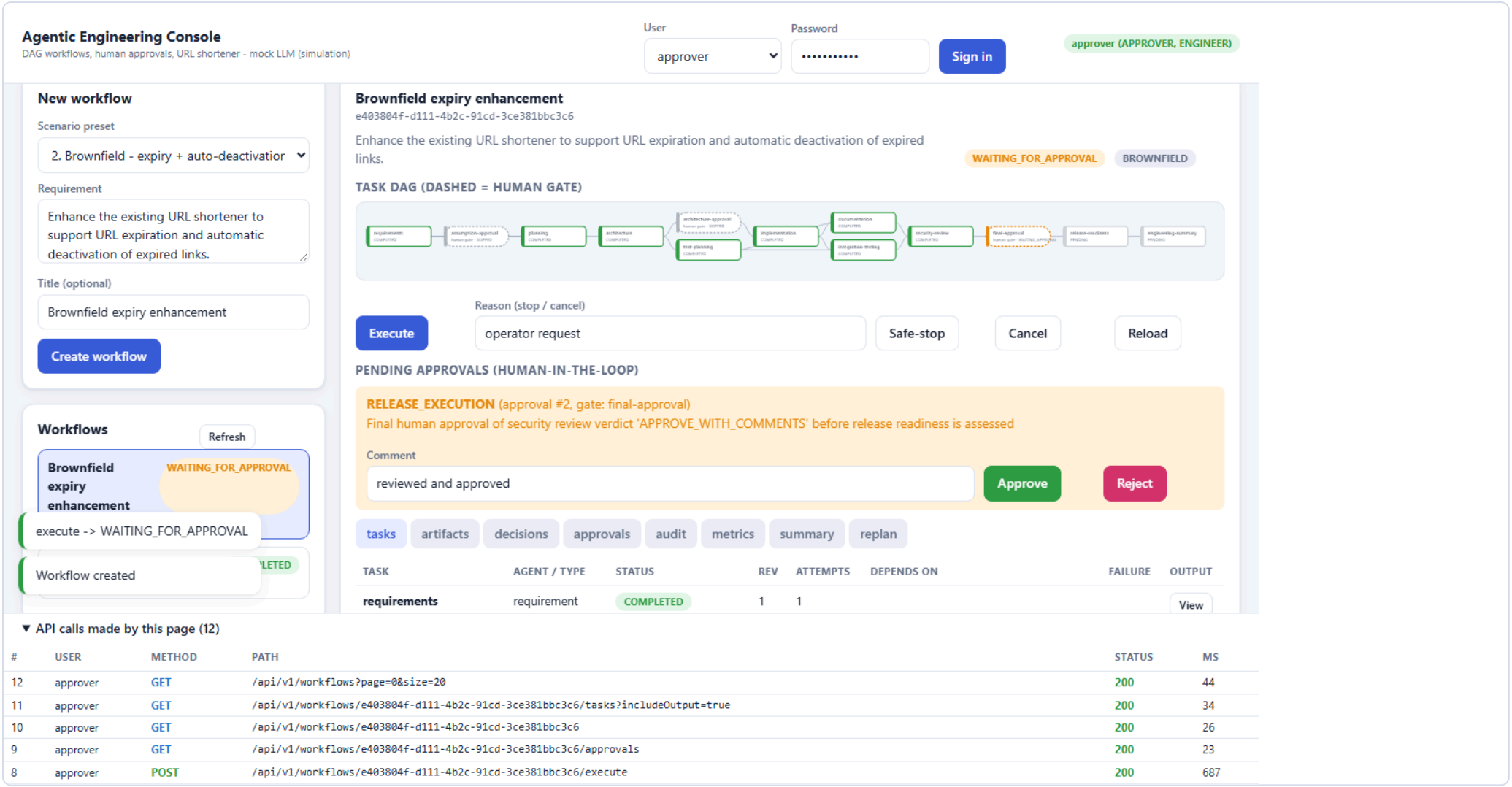


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/a4aa1d35-d071-45db-bb39-f2e3e5a81dbf/summary	200	33

SCENARIO 2 - BROWNFIELD

Step 18. Brownfield workflow executed

Same DAG, different content. The architecture stage first scans the real URL shortener module of this application by reflection.



▼ API calls made by this page (12)

#	USER	METHOD	PATH	STATUS	MS
12	approver	GET	/api/v1/workflows?page=0&size=20	200	44
11	approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/tasks?includeOutput=true	200	34
10	approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6	200	26
9	approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/approvals	200	23
8	approver	POST	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/execute	200	687

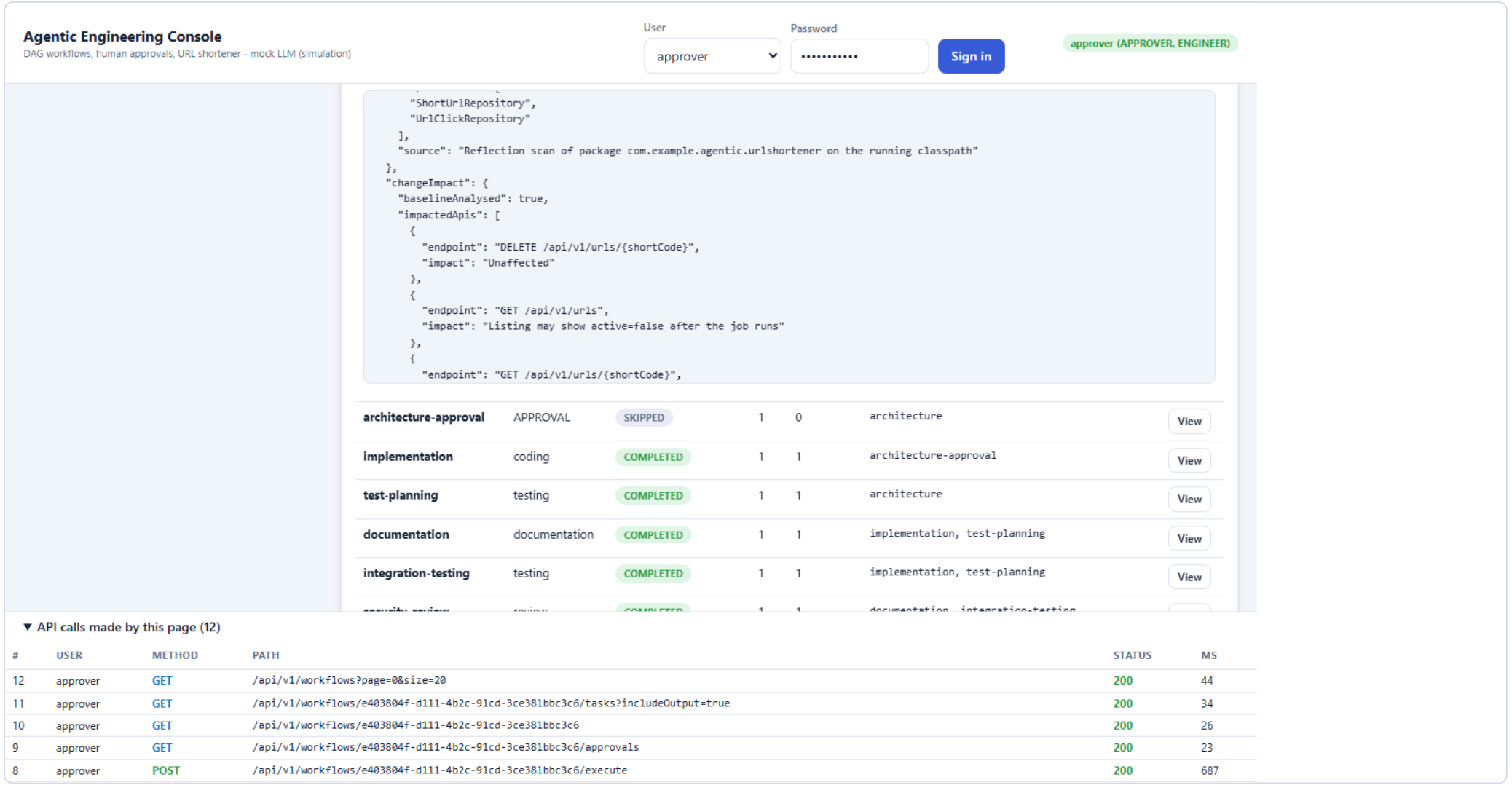
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	11
approver	POST	/api/v1/workflows	201	35

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	22
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/approvals	200	14
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6	200	18
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/tasks?includeOutput=true	200	19
approver	GET	/api/v1/workflows?page=0&size=20	200	26
approver	POST	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/execute	200	687
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/approvals	200	23
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6	200	26
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/tasks?includeOutput=true	200	34
approver	GET	/api/v1/workflows?page=0&size=20	200	44

SCENARIO 2 - BROWNFIELD

Step 19. Change impact analysis

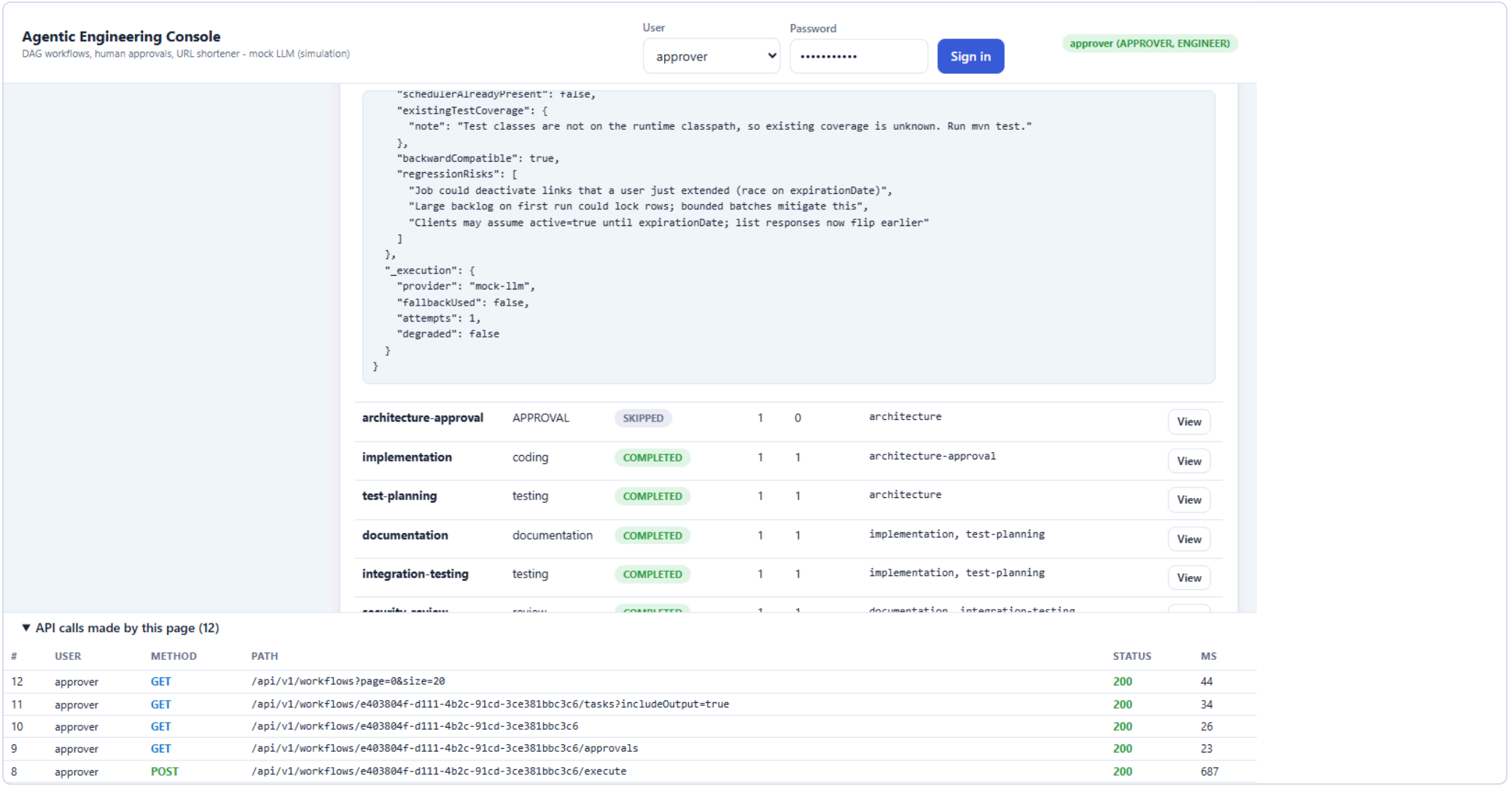
Architecture output: impacted APIs, entities, services and repositories discovered from the actual module, existing test discovery and backward-compatibility assessment.



SCENARIO 2 - BROWNFIELD

Step 20. Regression risks

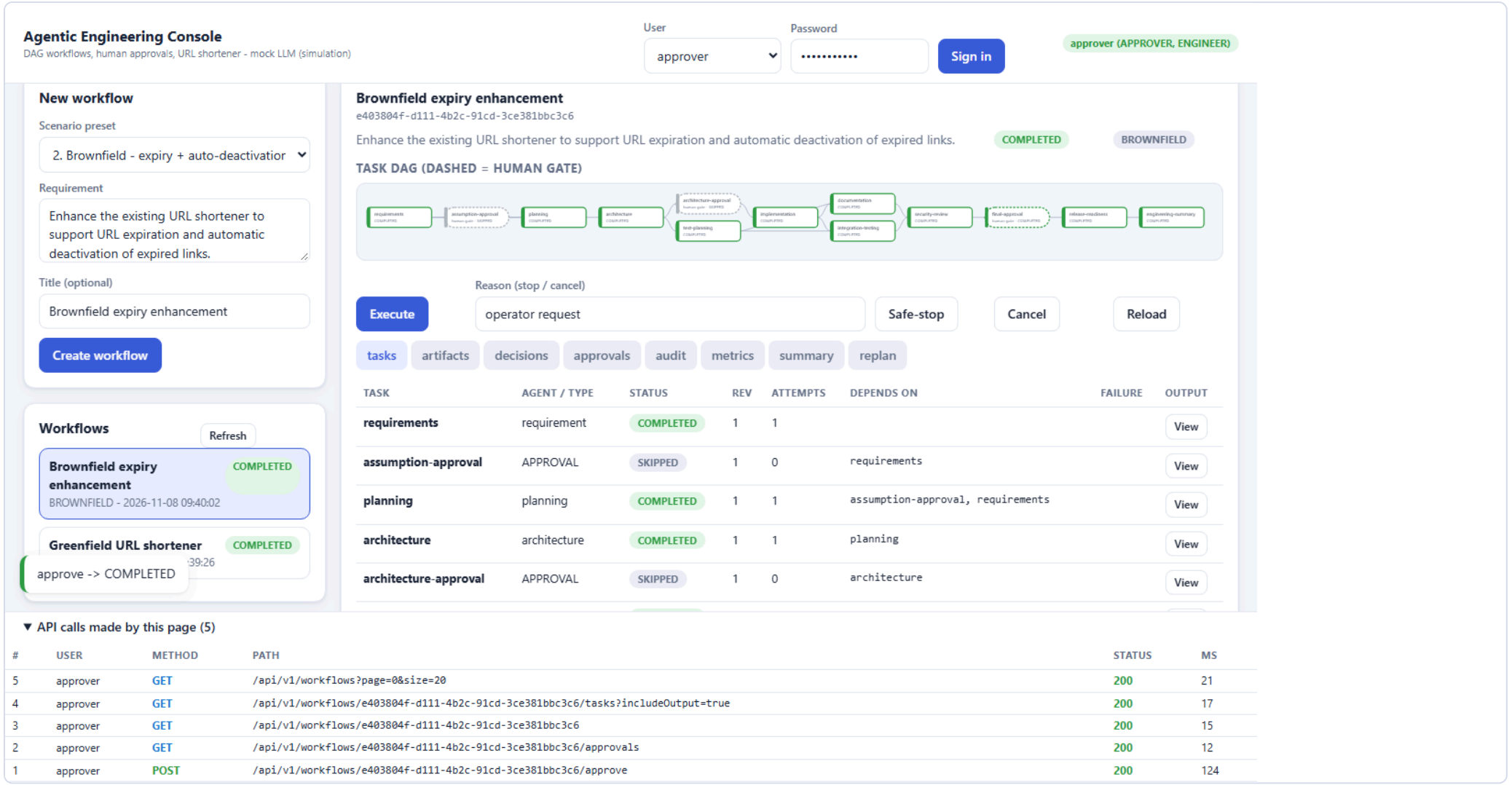
Regression risks and the statement that the change is backward compatible.



SCENARIO 2 - BROWNFIELD

Step 21. Brownfield completed

Approved and completed.



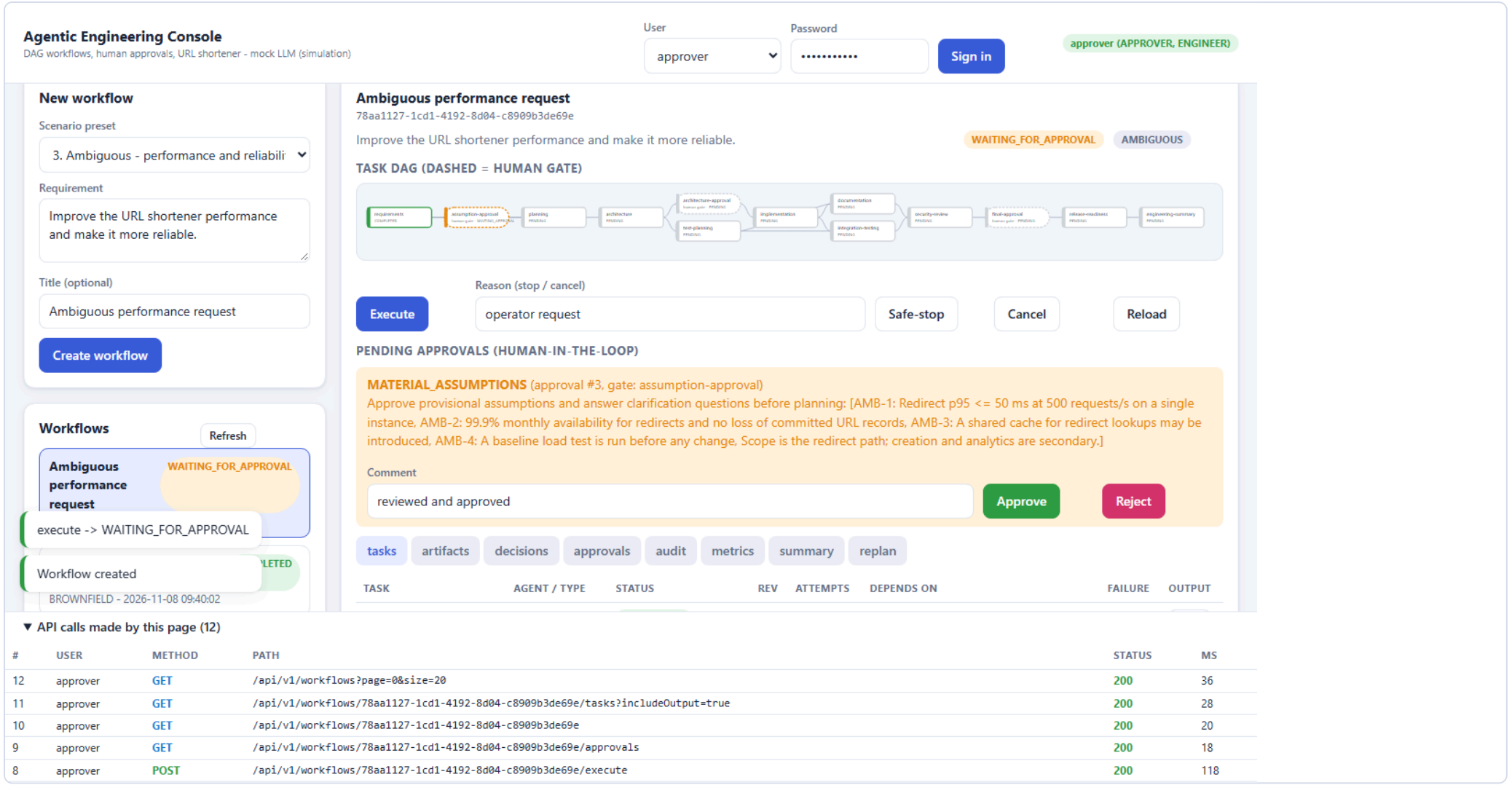
USER	METHOD	API CALL	HTTP	MS
approver	POST	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/approve	200	124
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/approvals	200	12

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6	200	15
approver	GET	/api/v1/workflows/e403804f-d111-4b2c-91cd-3ce381bbc3c6/tasks?includeOutput=true	200	17
approver	GET	/api/v1/workflows?page=0&size=20	200	21

SCENARIO 3 - AMBIGUOUS REQUIREMENT

Step 22. Paused for assumption approval

Blocking ambiguities were found. Planning has NOT started; the workflow waits for a human to approve the provisional assumptions.



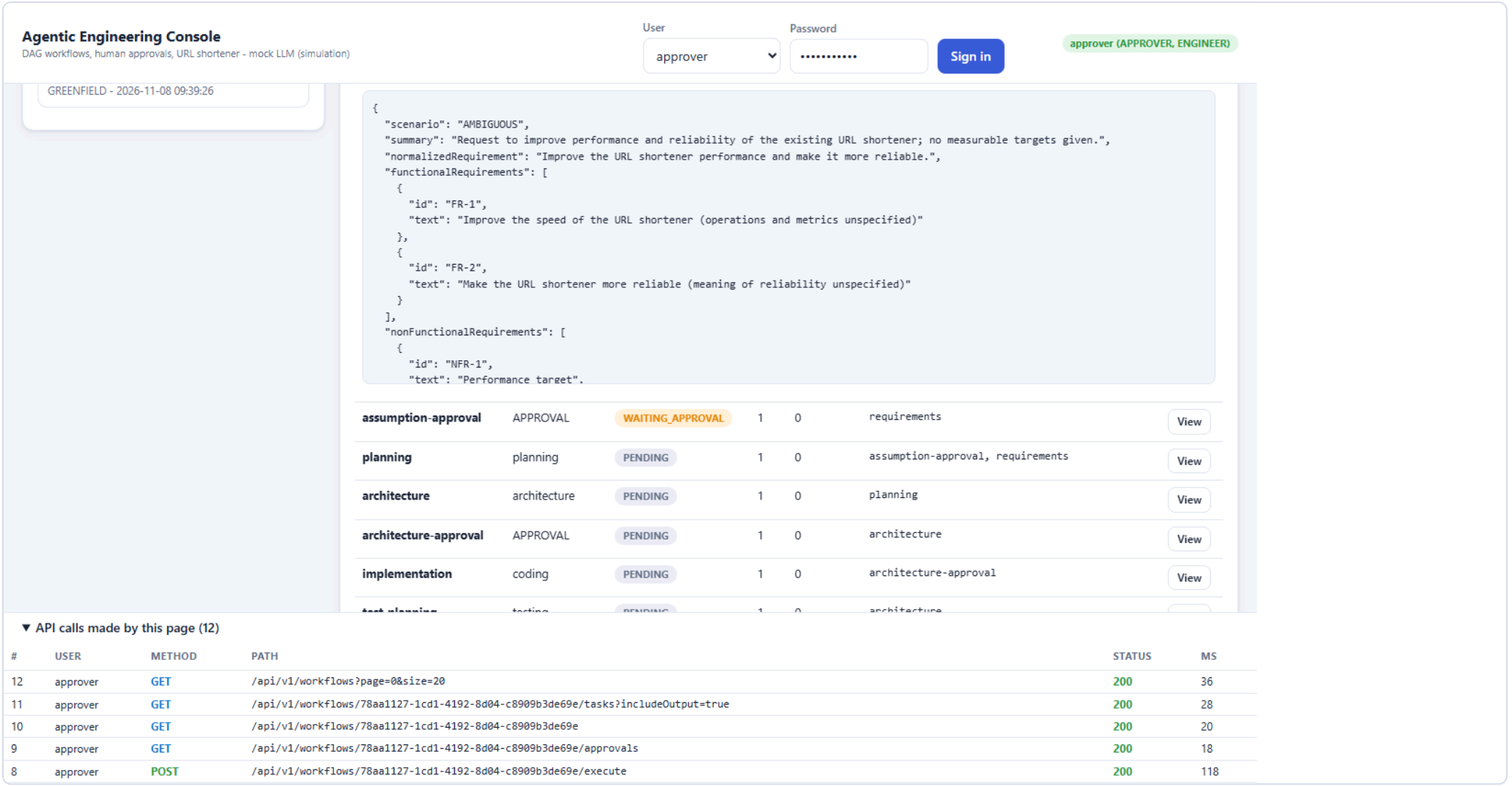
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	14
approver	POST	/api/v1/workflows	201	34

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	22
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	10
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	18
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	18
approver	GET	/api/v1/workflows?page=0&size=20	200	32
approver	POST	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/execute	200	118
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	18
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	20
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	28
approver	GET	/api/v1/workflows?page=0&size=20	200	36

SCENARIO 3 - AMBIGUOUS REQUIREMENT

Step 23. Questions and assumptions

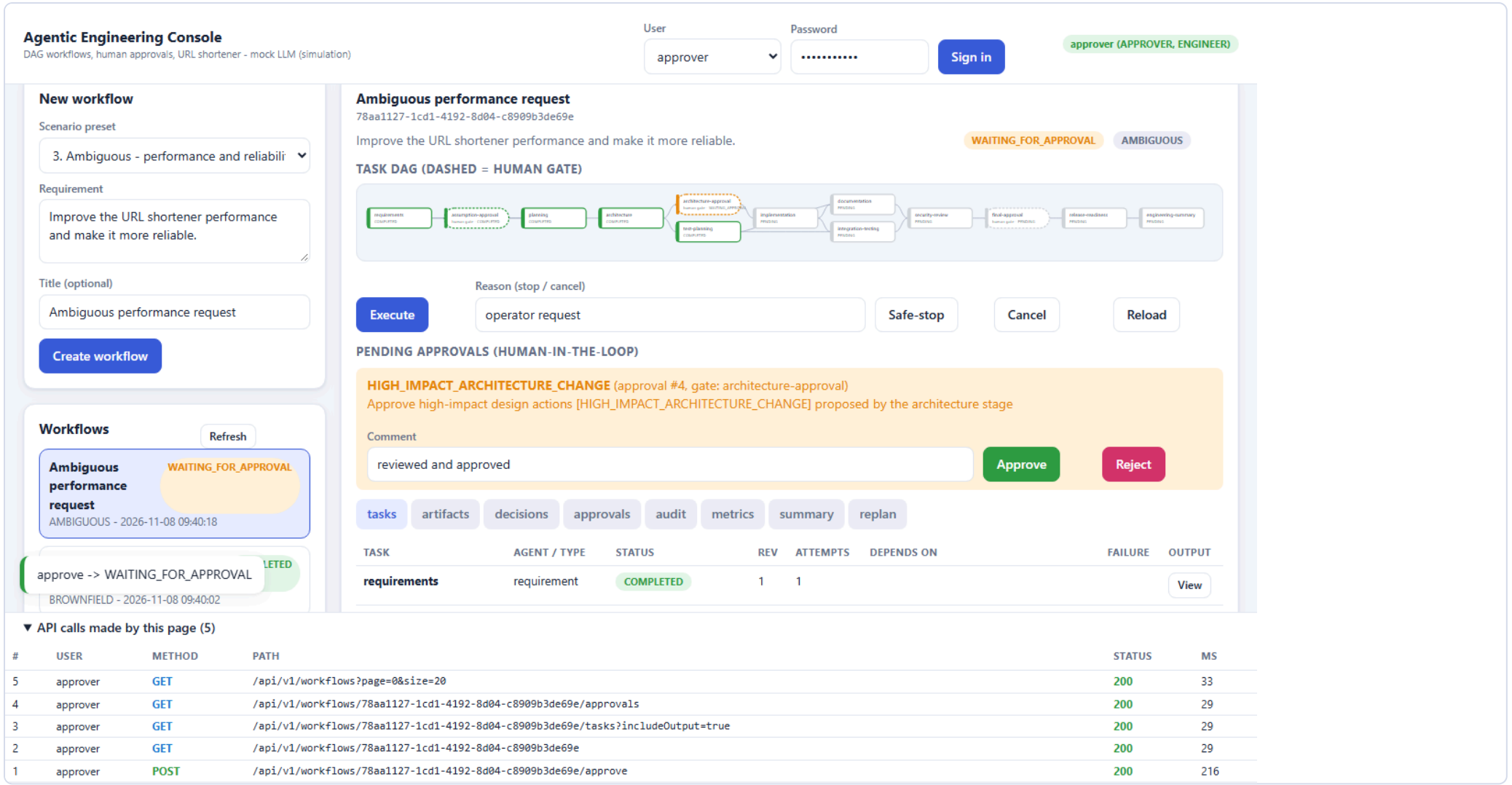
Requirements output: clarification questions, provisional assumptions (for example p95 latency and availability targets) and missing information such as baselines.



SCENARIO 3 - AMBIGUOUS REQUIREMENT

Step 24. Second gate: high-impact design

After the assumptions were approved the plan was produced. Introducing a shared cache is a high-impact architecture change, so a second approval is required. test-planning already completed in parallel.



▼ API calls made by this page (5)

#	USER	METHOD	PATH	STATUS	MS
5	approver	GET	/api/v1/workflows?page=0&size=20	200	33
4	approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	29
3	approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	29
2	approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	29
1	approver	POST	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approve	200	216

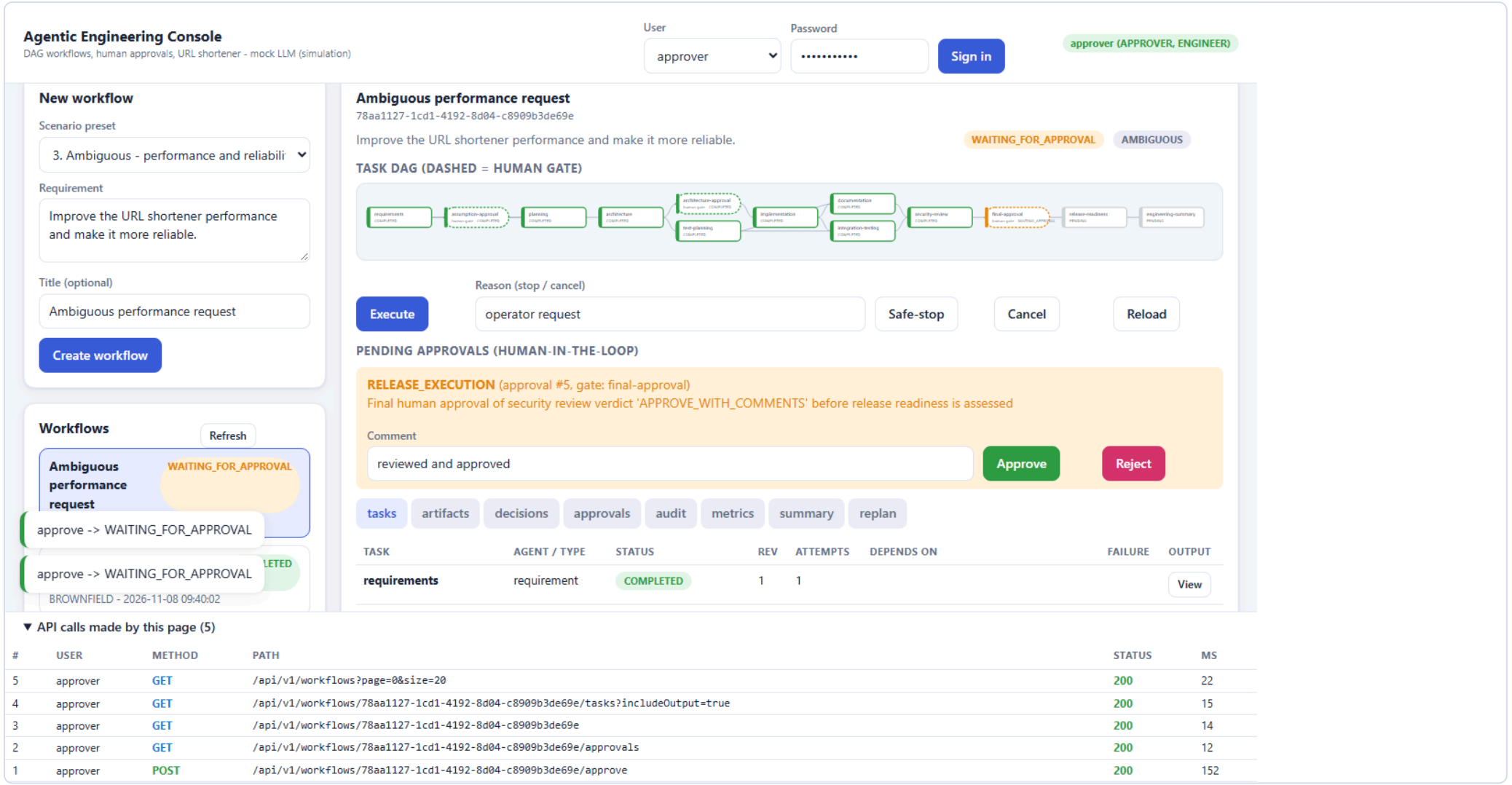
USER	METHOD	API CALL	HTTP	MS
approver	POST	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approve	200	216
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	29

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	29
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	29
approver	GET	/api/v1/workflows?page=0&size=20	200	33

SCENARIO 3 - AMBIGUOUS REQUIREMENT

Step 25. Third gate: release

Design approved; implementation, documentation, validation and review ran; final release approval pending.



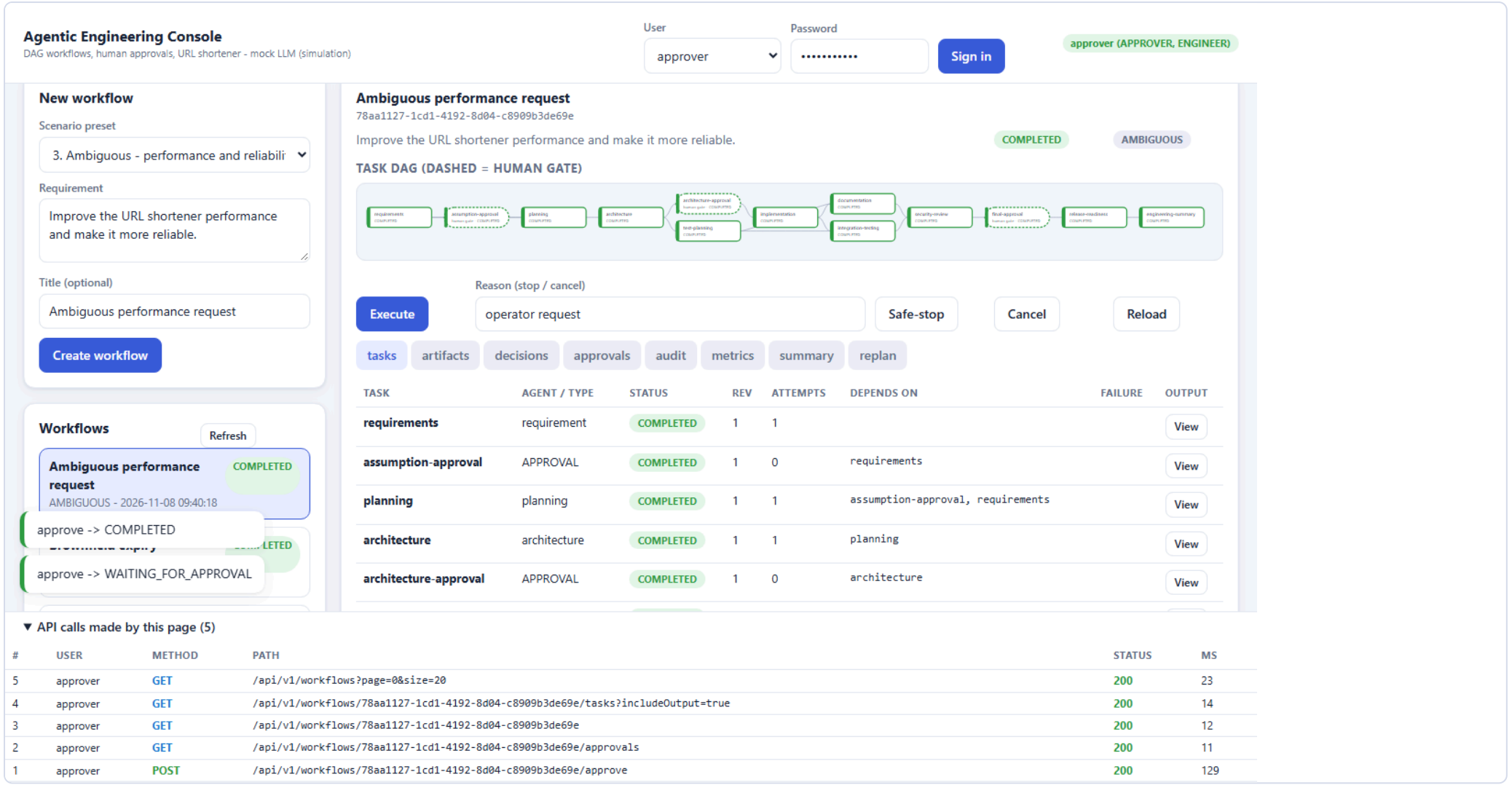
USER	METHOD	API CALL	HTTP	MS
approver	POST	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approve	200	152
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	12

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	14
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	15
approver	GET	/api/v1/workflows?page=0&size=20	200	22

SCENARIO 3 - AMBIGUOUS REQUIREMENT

Step 26. Ambiguous scenario completed

Completed after three human approvals. Release readiness is READY_WITH_CONDITIONS because the targets are still provisional.



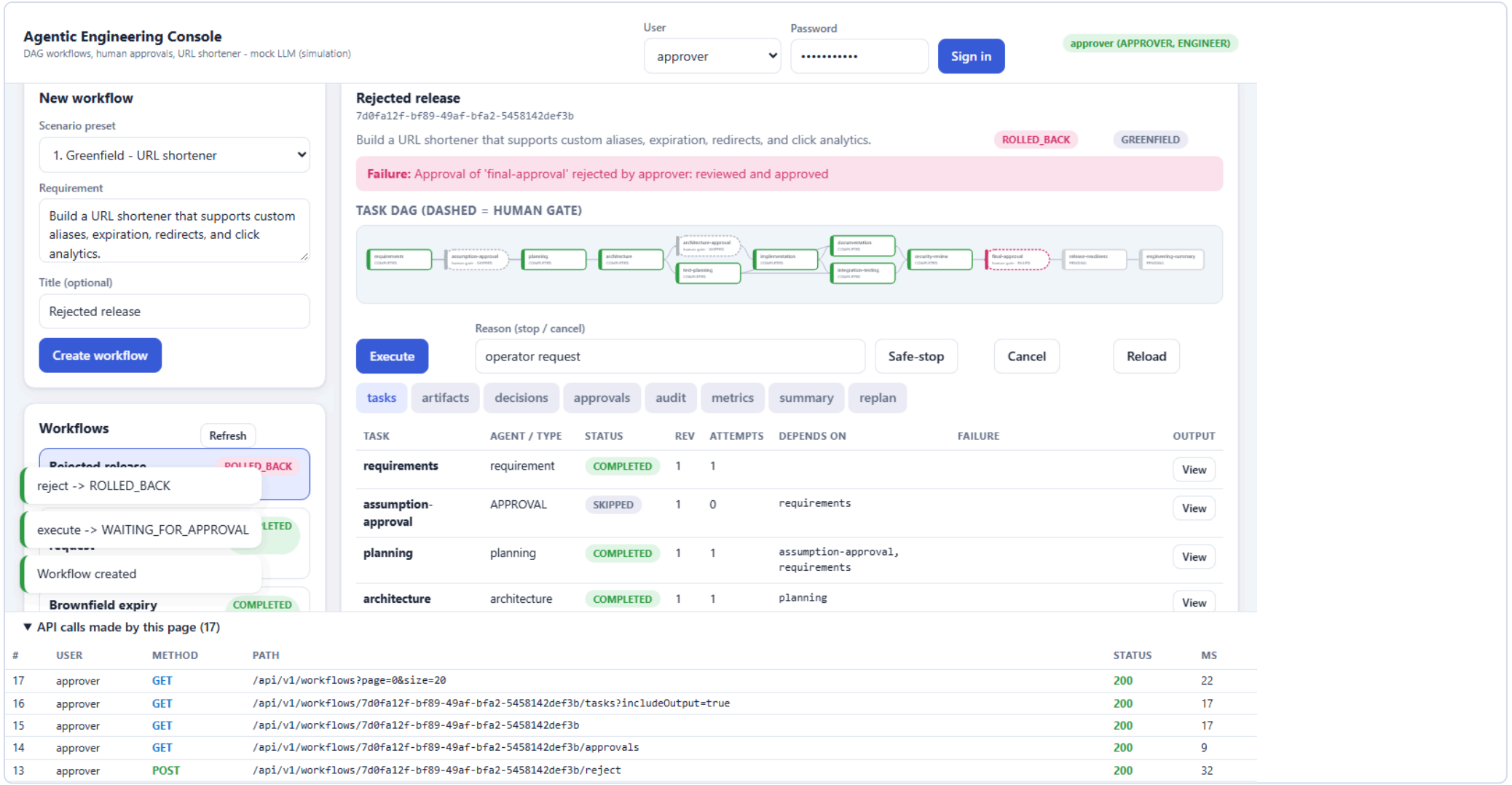
USER	METHOD	API CALL	HTTP	MS
approver	POST	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approve	200	129
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/approvals	200	11

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e	200	12
approver	GET	/api/v1/workflows/78aa1127-1cd1-4192-8d04-c8909b3de69e/tasks?includeOutput=true	200	14
approver	GET	/api/v1/workflows?page=0&size=20	200	23

REJECTION AND ROLLBACK

Step 27. Rejected, rolled back

POST /reject: the workflow is ROLLED_BACK and its staged artifact versions are reverted (a safe, reversible operation).



USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	25
approver	POST	/api/v1/workflows	201	26

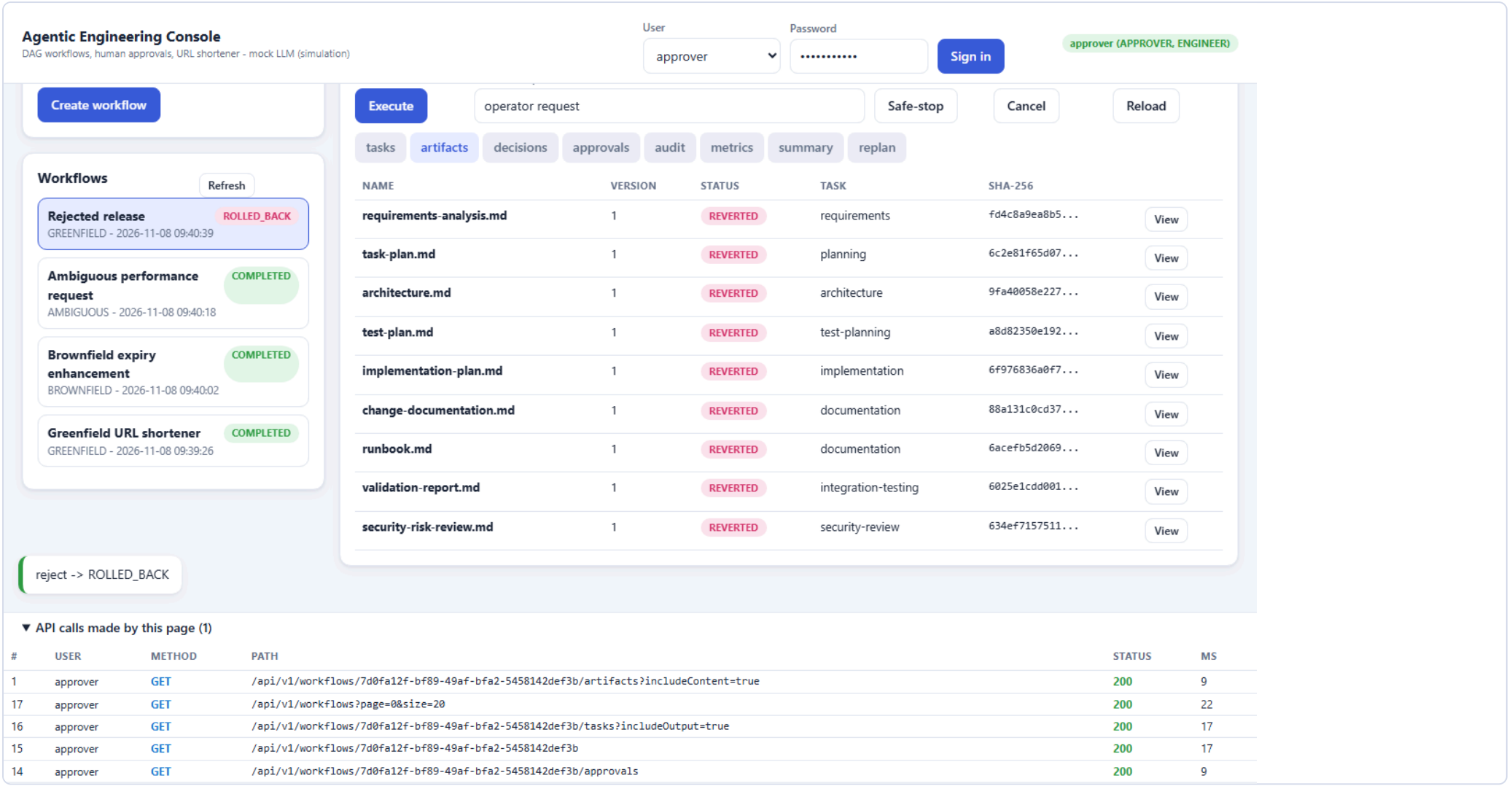
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	19
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/approvals	200	13
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b	200	28
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/tasks?includeOutput=true	200	29
approver	GET	/api/v1/workflows?page=0&size=20	200	23
approver	POST	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/execute	200	201
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/approvals	200	10
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b	200	19
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/tasks?includeOutput=true	200	20
approver	GET	/api/v1/workflows?page=0&size=20	200	27
approver	POST	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/reject	200	32
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/approvals	200	9
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b	200	17
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/tasks?includeOutput=true	200	17

... and 1 more calls in this step

REJECTION AND ROLLBACK

Step 28. Artifacts reverted

All artifacts are REVERTED; nothing was accepted.

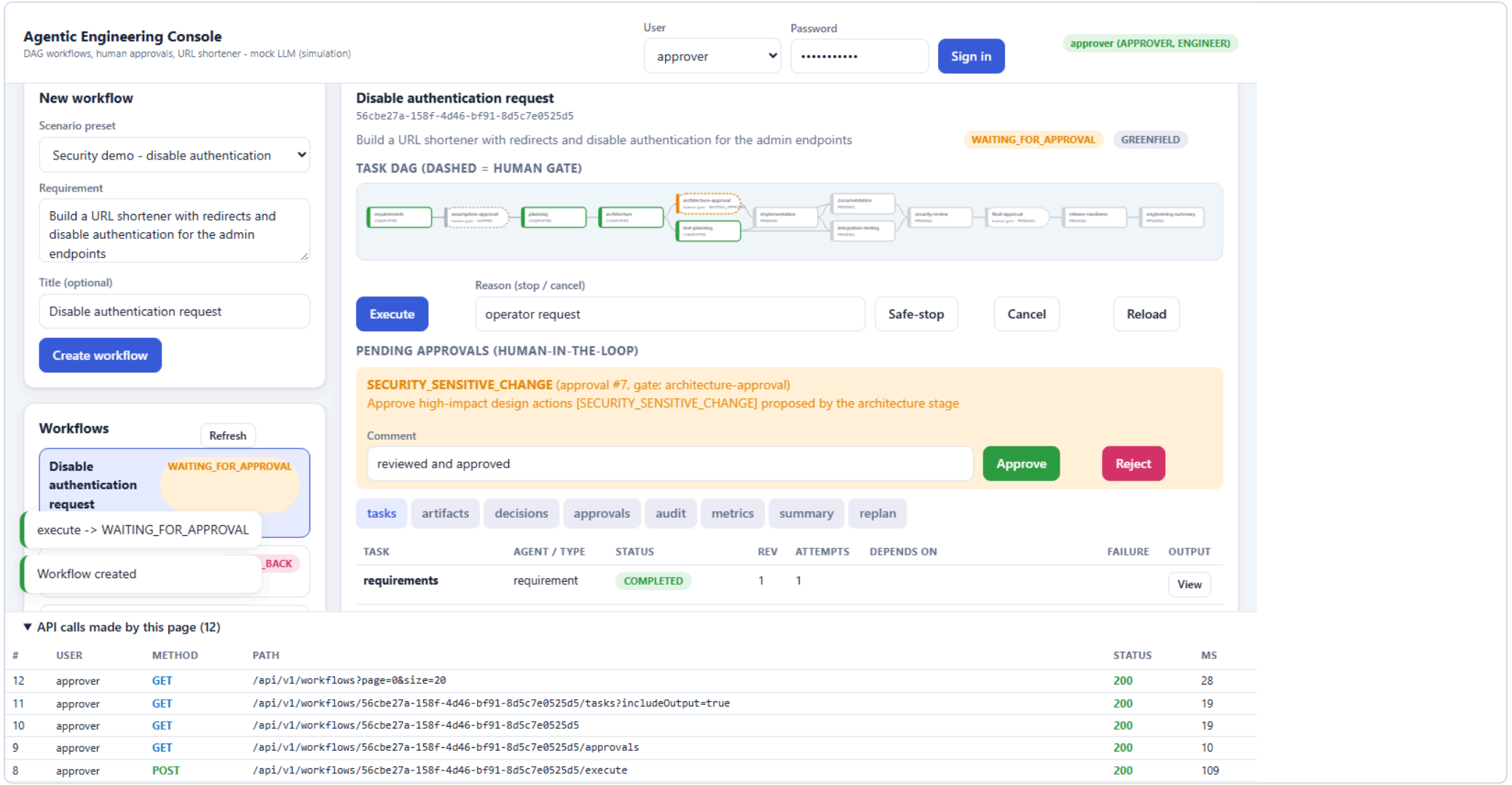


USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/7d0fa12f-bf89-49af-bfa2-5458142def3b/artifacts?includeContent=true	200	9

SAFE-STOP (SECURITY)

Step 29. Design approval for a sensitive change

The architecture marks a SECURITY_SENSITIVE_CHANGE, which needs a human approval before implementation.



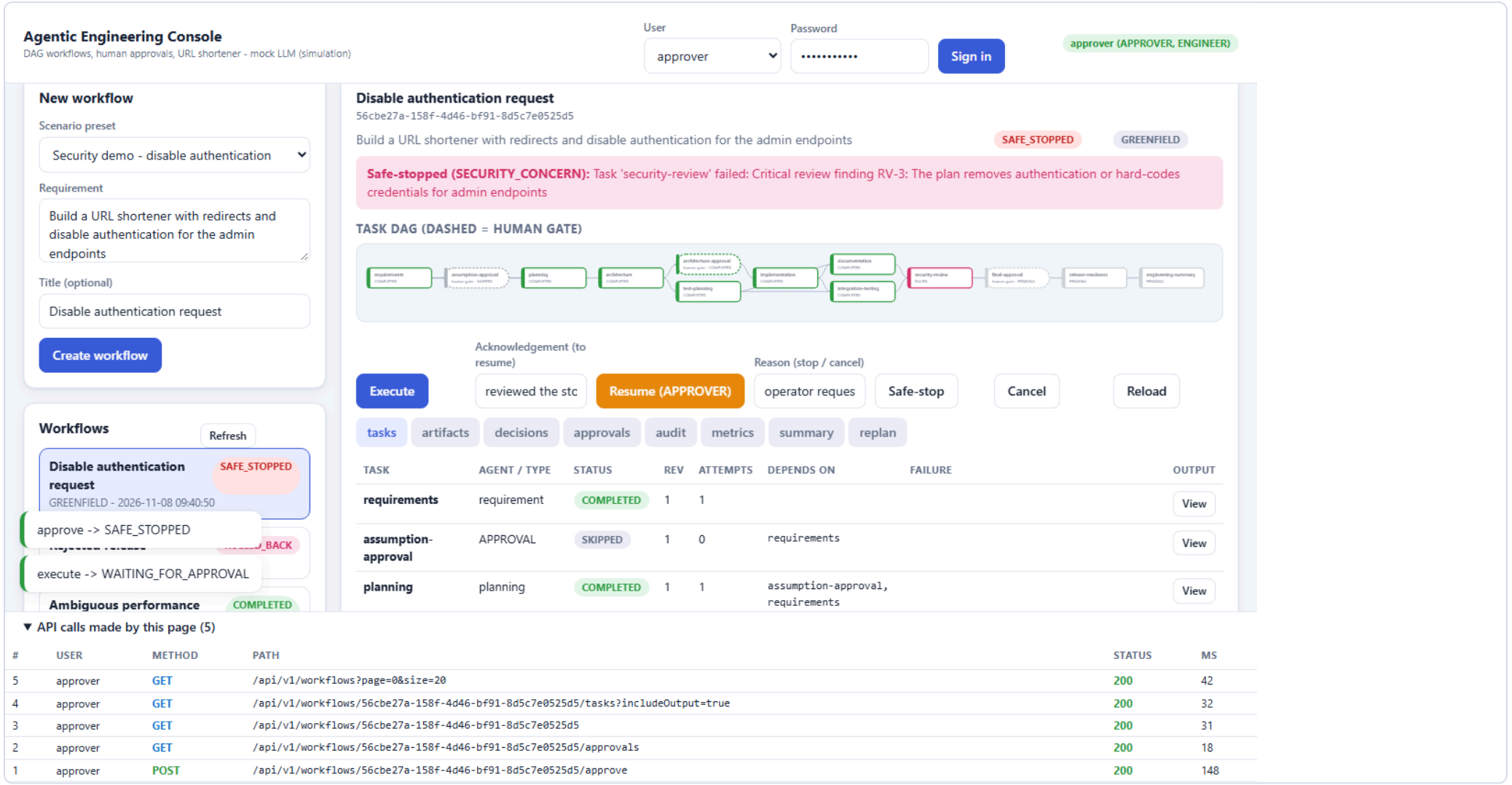
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	16
approver	POST	/api/v1/workflows	201	41

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=20	200	33
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	27
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	27
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	27
approver	GET	/api/v1/workflows?page=0&size=20	200	47
approver	POST	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/execute	200	109
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	10
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	19
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	19
approver	GET	/api/v1/workflows?page=0&size=20	200	28

SAFE-STOP (SECURITY)

Step 30. Safe-stopped on a critical finding

The review stage found a critical security finding and the workflow stopped (SAFE_STOPPED, SECURITY_CONCERN). The reason is persisted; nothing downstream ran.



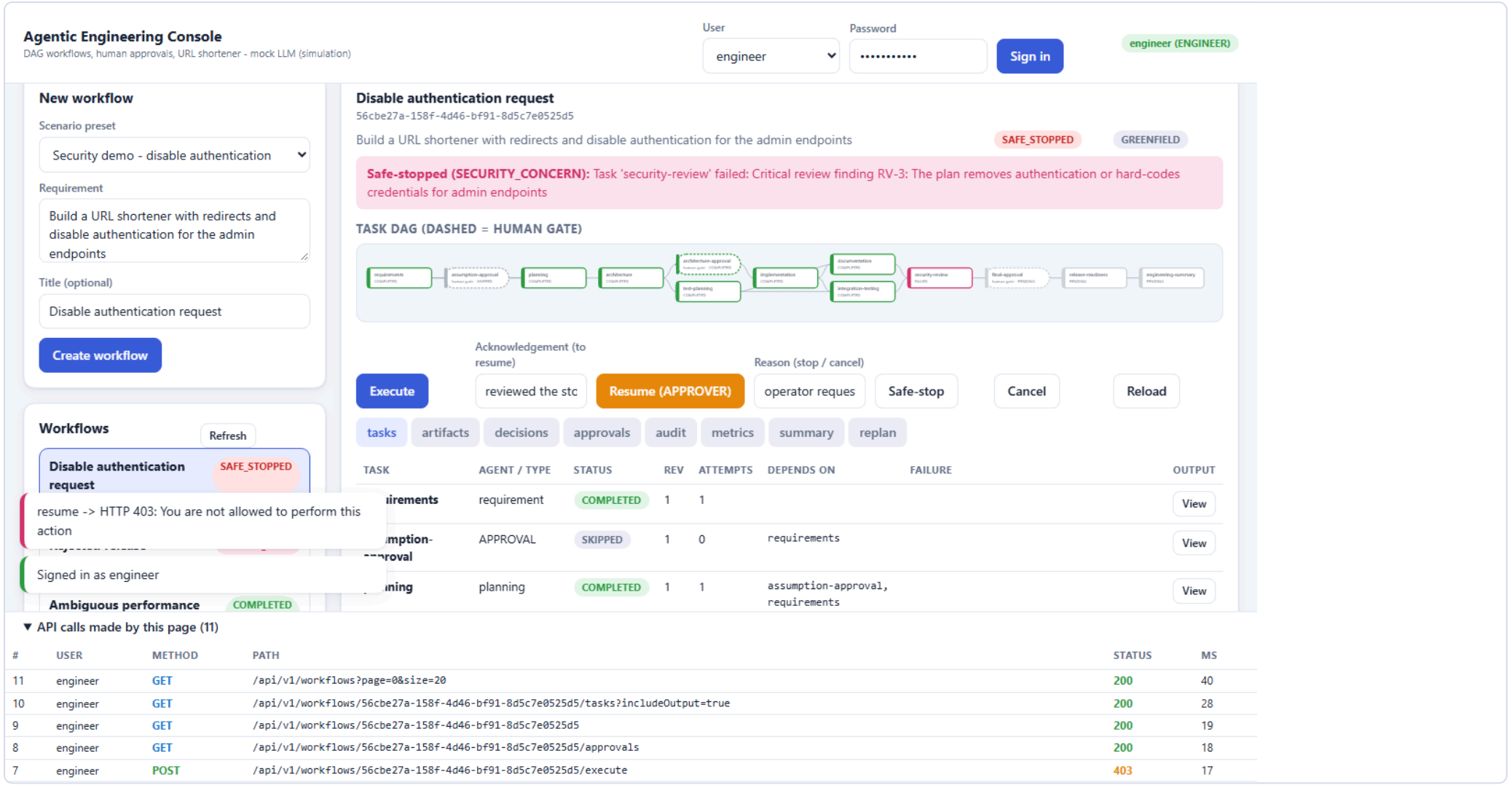
USER	METHOD	API CALL	HTTP	MS
approver	POST	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approve	200	148
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	18

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	31
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	32
approver	GET	/api/v1/workflows?page=0&size=20	200	42

SAFE-STOP (SECURITY)

Step 31. Engineer cannot resume (403)

Resuming a safe-stopped workflow requires the APPROVER role and an acknowledgement.



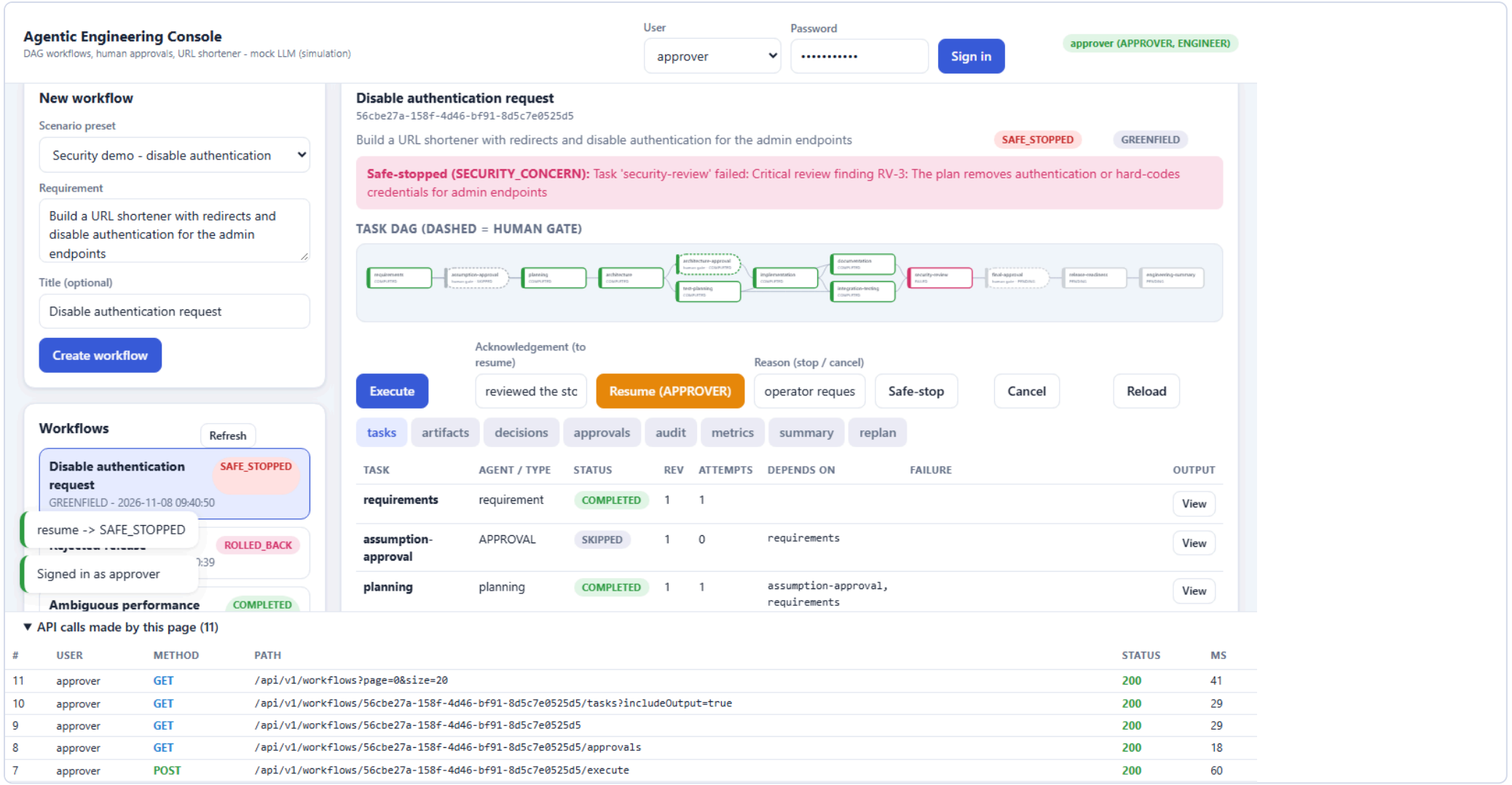
USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/whoami	200	5
engineer	GET	/api/v1/workflows?page=0&size=20	200	18

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	9
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	14
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	14
engineer	GET	/api/v1/workflows?page=0&size=20	200	23
engineer	POST	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/execute	403	17
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	18
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	19
engineer	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	28
engineer	GET	/api/v1/workflows?page=0&size=20	200	40

SAFE-STOP (SECURITY)

Step 32. Approver resumes: stopped again

The approver resumed with an acknowledgement; the same policy violation stops it again, proving the guard cannot be bypassed.



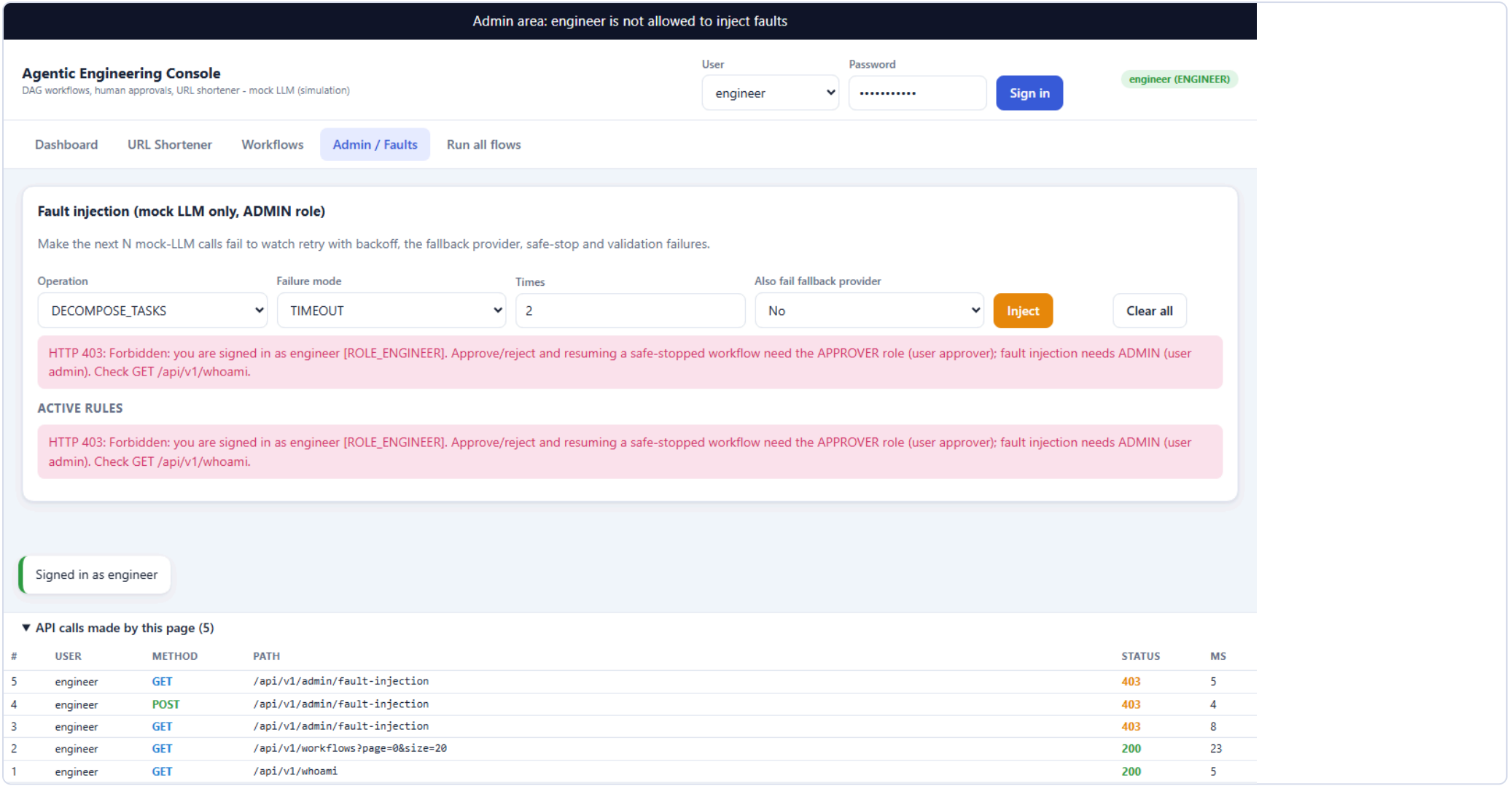
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/whoami	200	5
approver	GET	/api/v1/workflows?page=0&size=20	200	16

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	17
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	27
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	27
approver	GET	/api/v1/workflows?page=0&size=20	200	29
approver	POST	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/execute	200	60
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/approvals	200	18
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5	200	29
approver	GET	/api/v1/workflows/56cbe27a-158f-4d46-bf91-8d5c7e0525d5/tasks?includeOutput=true	200	29
approver	GET	/api/v1/workflows?page=0&size=20	200	41

RETRY, FALLBACK AND FAULT INJECTION

Step 33. Fault injection denied for engineer

POST /api/v1/admin/fault-injection returned 403; the message names the required ADMIN role.



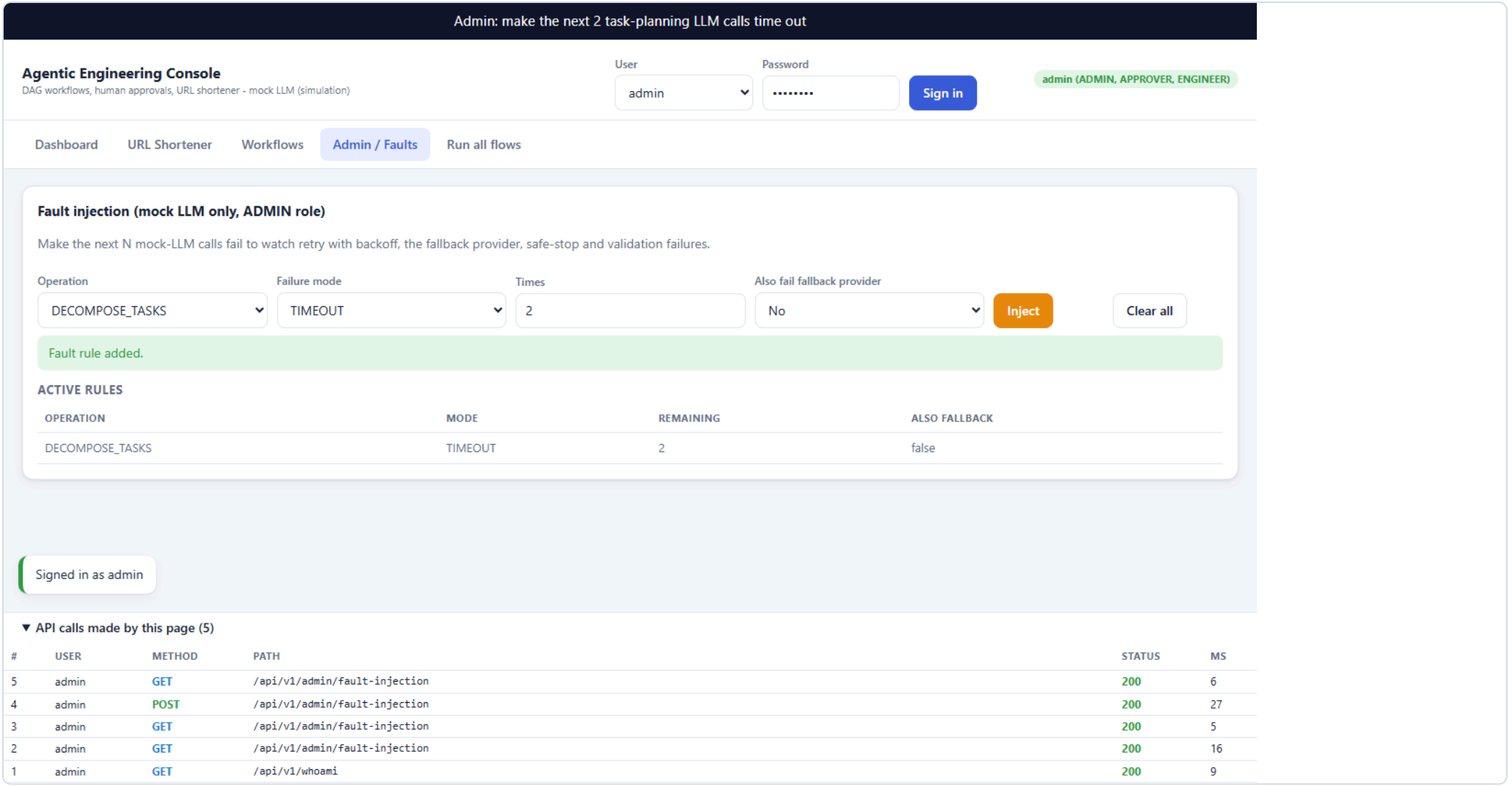
USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/whoami	200	5
engineer	GET	/api/v1/workflows?page=0&size=20	200	23

USER	METHOD	API CALL	HTTP	MS
engineer	GET	/api/v1/admin/fault-injection	403	8
engineer	POST	/api/v1/admin/fault-injection	403	4
engineer	GET	/api/v1/admin/fault-injection	403	5

RETRY, FALLBACK AND FAULT INJECTION

Step 34. Fault rule injected

The next two DECOMPOSE_TASKS calls will fail with a timeout.



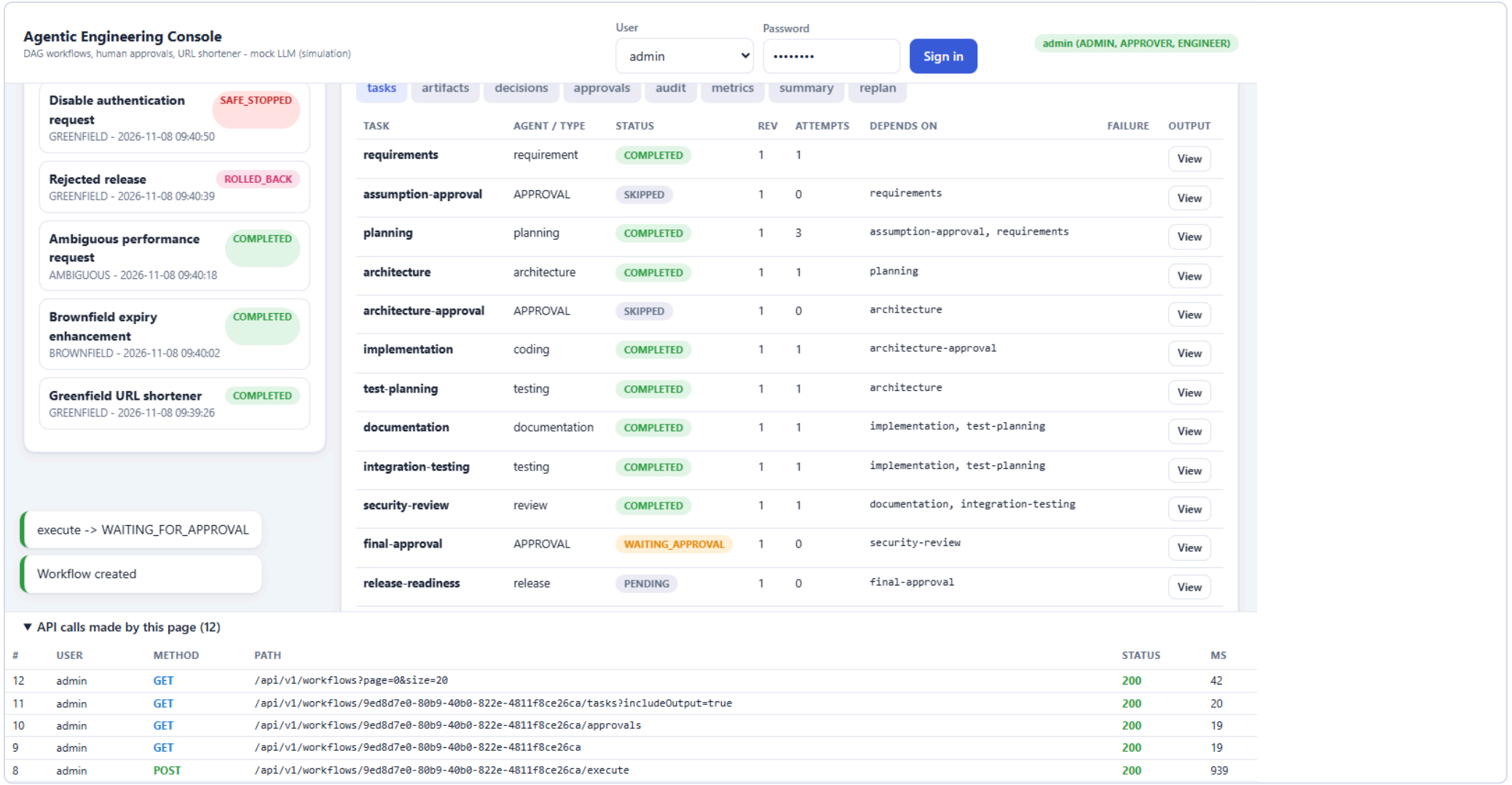
USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/whoami	200	9
admin	GET	/api/v1/admin/fault-injection	200	16

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/admin/fault-injection	200	5
admin	POST	/api/v1/admin/fault-injection	200	27
admin	GET	/api/v1/admin/fault-injection	200	6

RETRY, FALLBACK AND FAULT INJECTION

Step 35. Planning needed 3 attempts

The planning task shows Attempts = 3: two transient failures were retried and the third attempt succeeded.



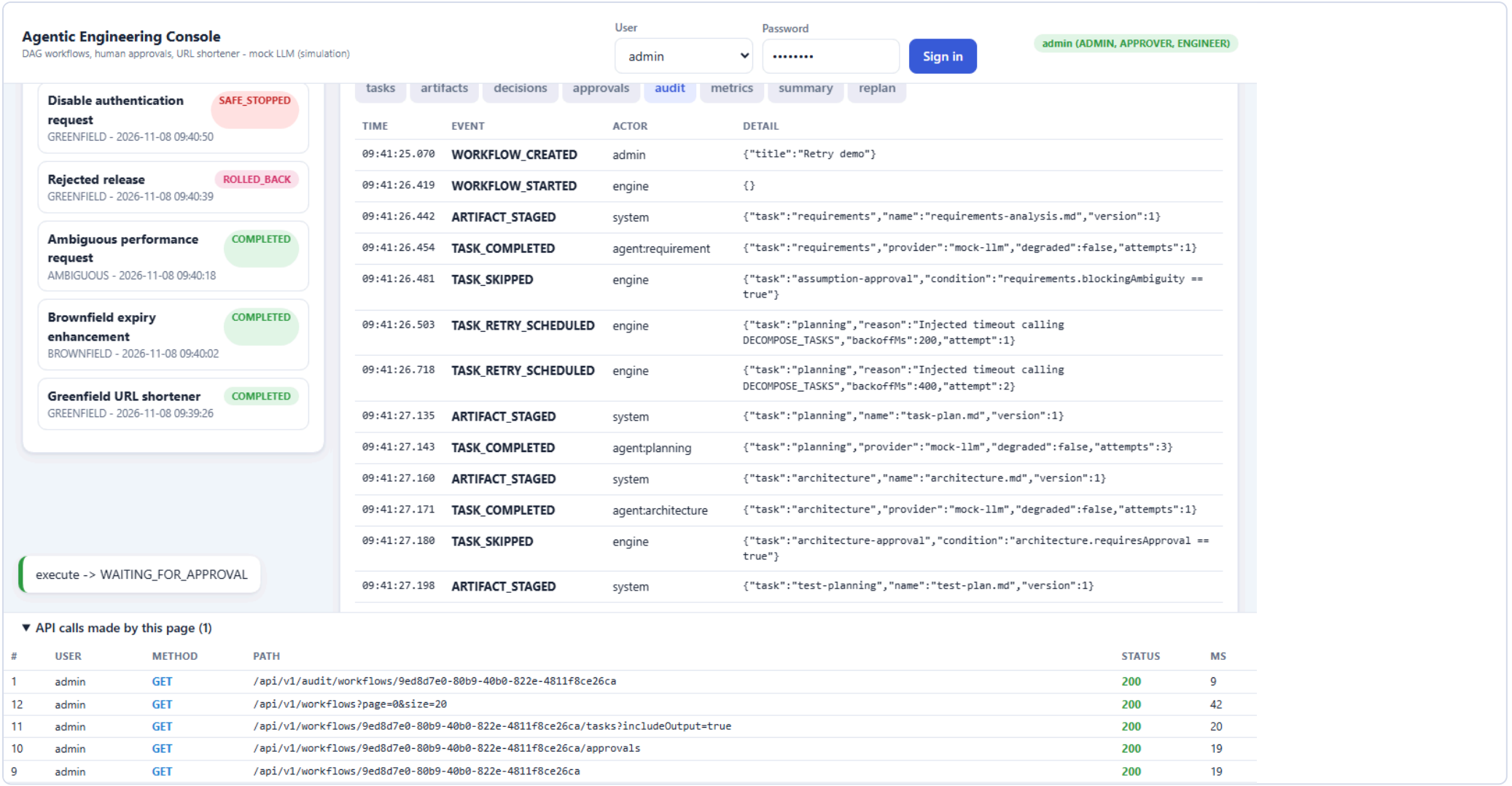
USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows?page=0&size=20	200	26
admin	POST	/api/v1/workflows	201	20

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows?page=0&size=20	200	23
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	11
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	11
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	11
admin	GET	/api/v1/workflows?page=0&size=20	200	30
admin	POST	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/execute	200	939
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	19
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	19
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	20
admin	GET	/api/v1/workflows?page=0&size=20	200	42

RETRY, FALLBACK AND FAULT INJECTION

Step 36. Retry audit events

TASK_RETRY_SCHEDULED events record each attempt and the backoff (200 ms, then 400 ms).

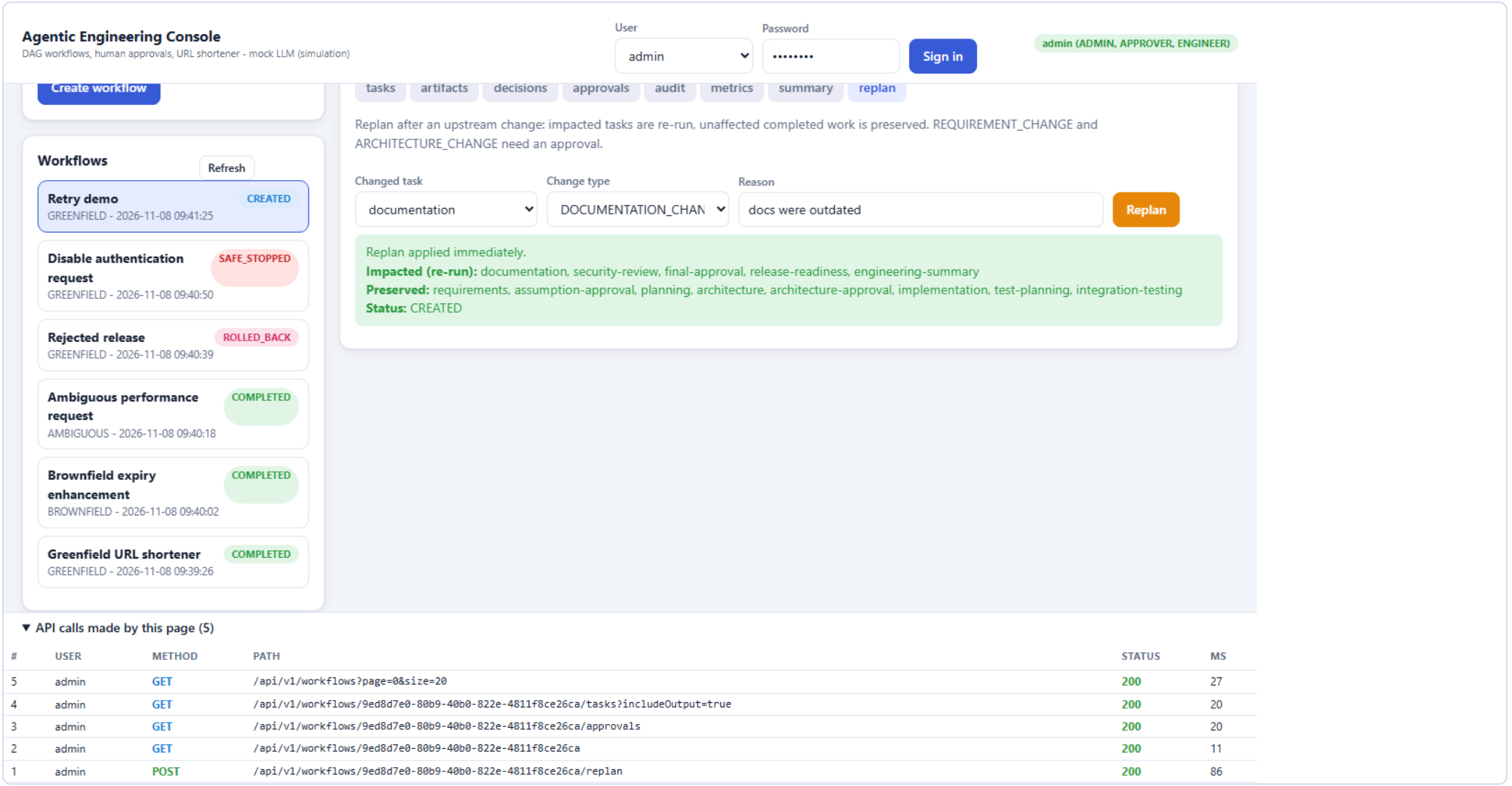


USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/audit/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	9

DYNAMIC REPLANNING

Step 37. Low-impact replan applied

POST /replan: only documentation and everything downstream are marked for re-execution; completed upstream work is preserved. No approval is needed for a low-impact change.



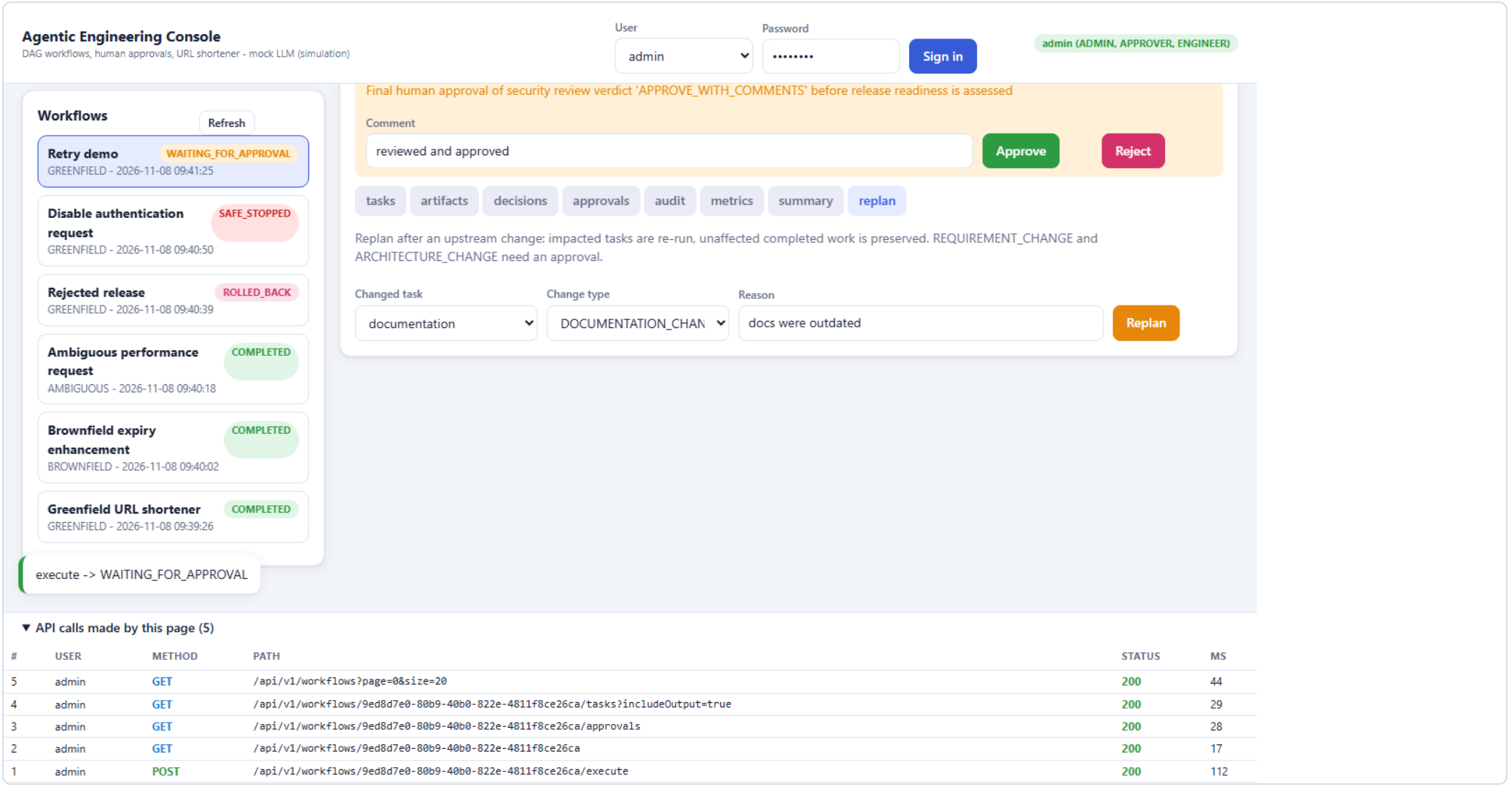
USER	METHOD	API CALL	HTTP	MS
admin	POST	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/replan	200	86
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	11

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	20
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	20
admin	GET	/api/v1/workflows?page=0&size=20	200	27

DYNAMIC REPLANNING

Step 38. Only stale tasks re-ran

The workflow re-ran just the affected tasks (revision 2) and paused again at the final approval.



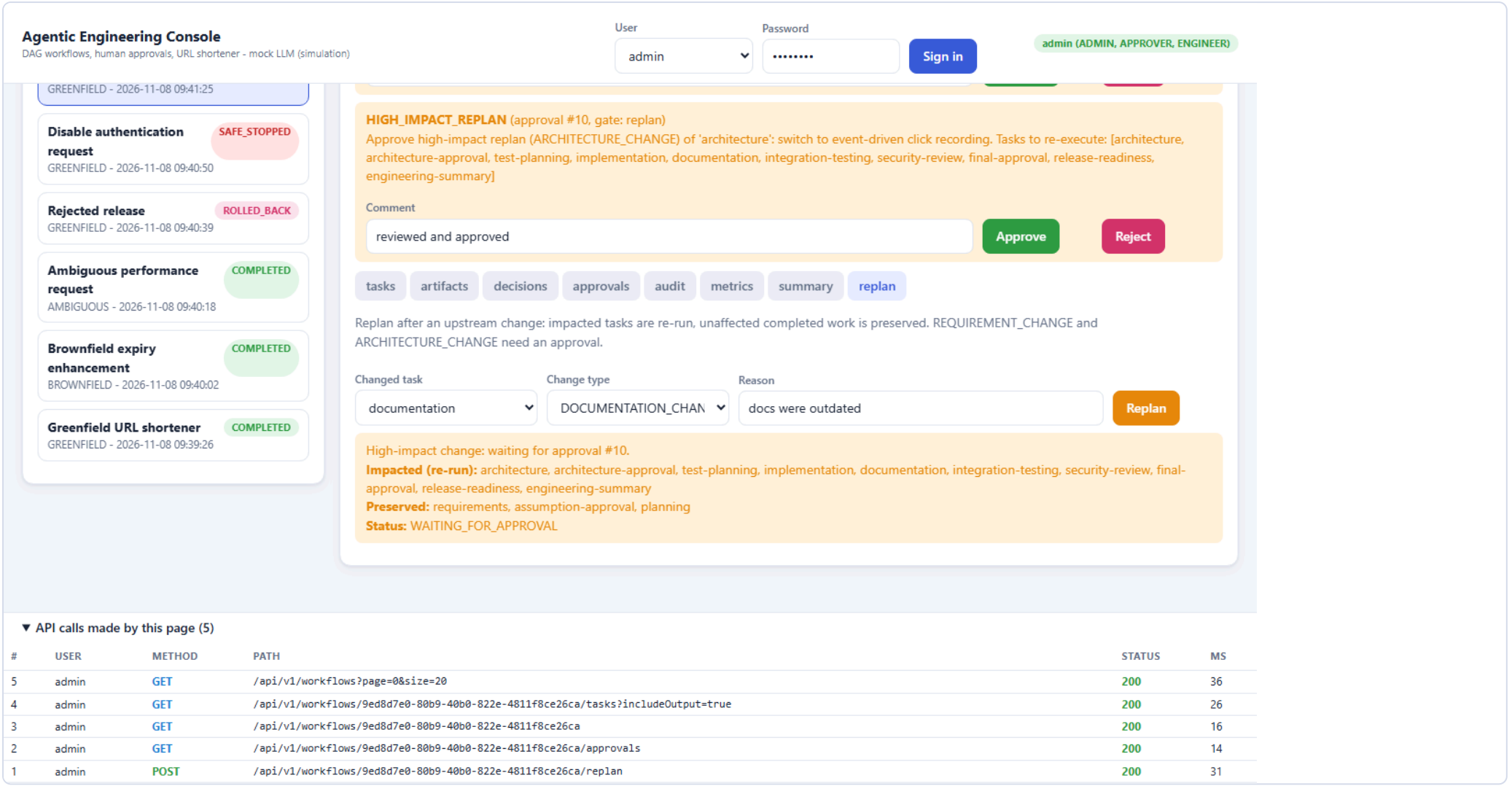
USER	METHOD	API CALL	HTTP	MS
admin	POST	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/execute	200	112
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	17

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	28
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	29
admin	GET	/api/v1/workflows?page=0&size=20	200	44

DYNAMIC REPLANNING

Step 39. High-impact replan waits for approval

Nothing changes until a human approves the proposal; the exact impact (impacted and preserved tasks) is shown for review.



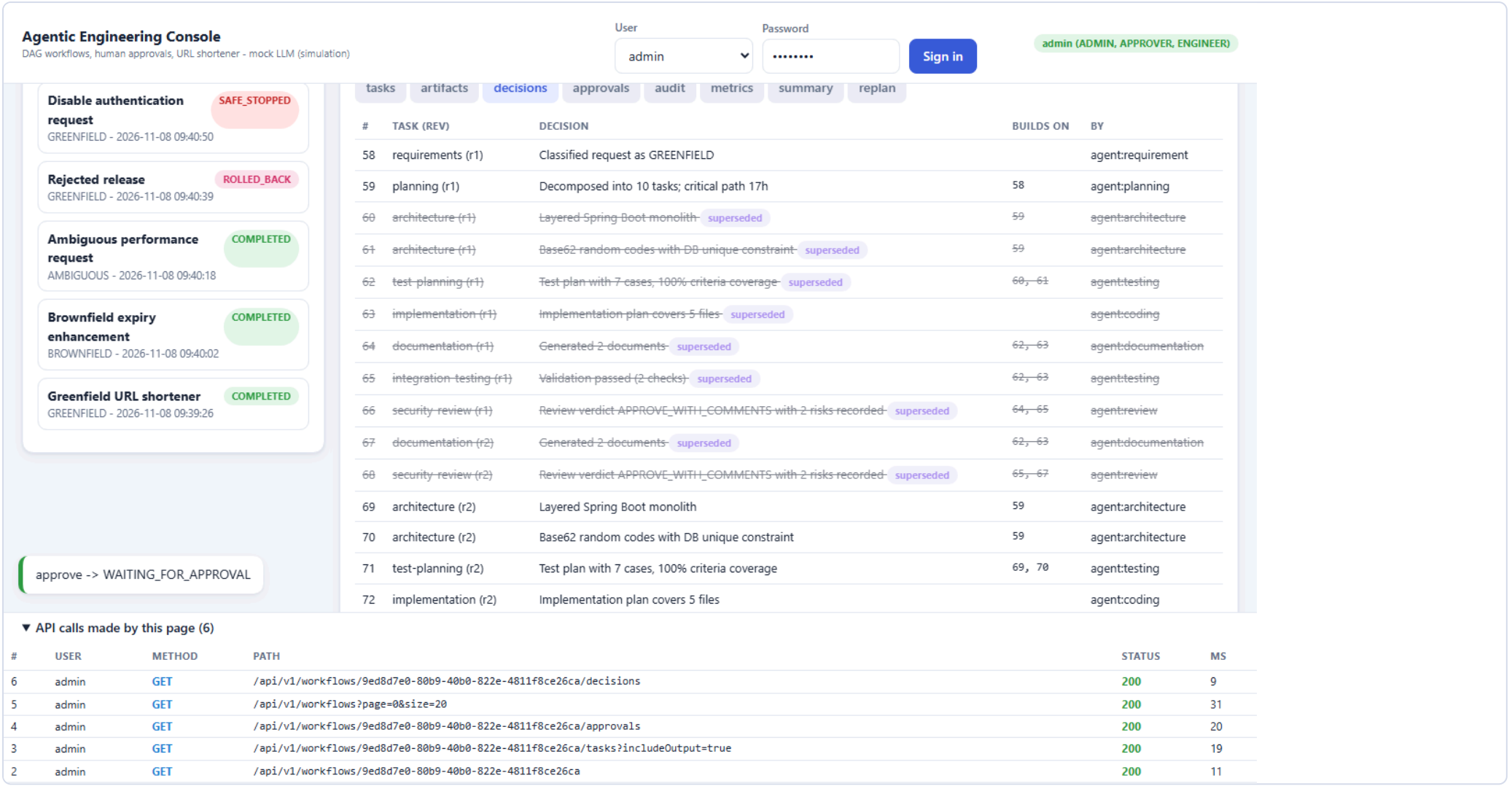
USER	METHOD	API CALL	HTTP	MS
admin	POST	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/replan	200	31
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	14

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	16
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	26
admin	GET	/api/v1/workflows?page=0&size=20	200	36

DYNAMIC REPLANNING

Step 40. Replan approved, decisions superseded

After approval the architecture and its dependants re-ran. Earlier decisions are struck through (superseded) but kept for traceability, never silently overwritten.



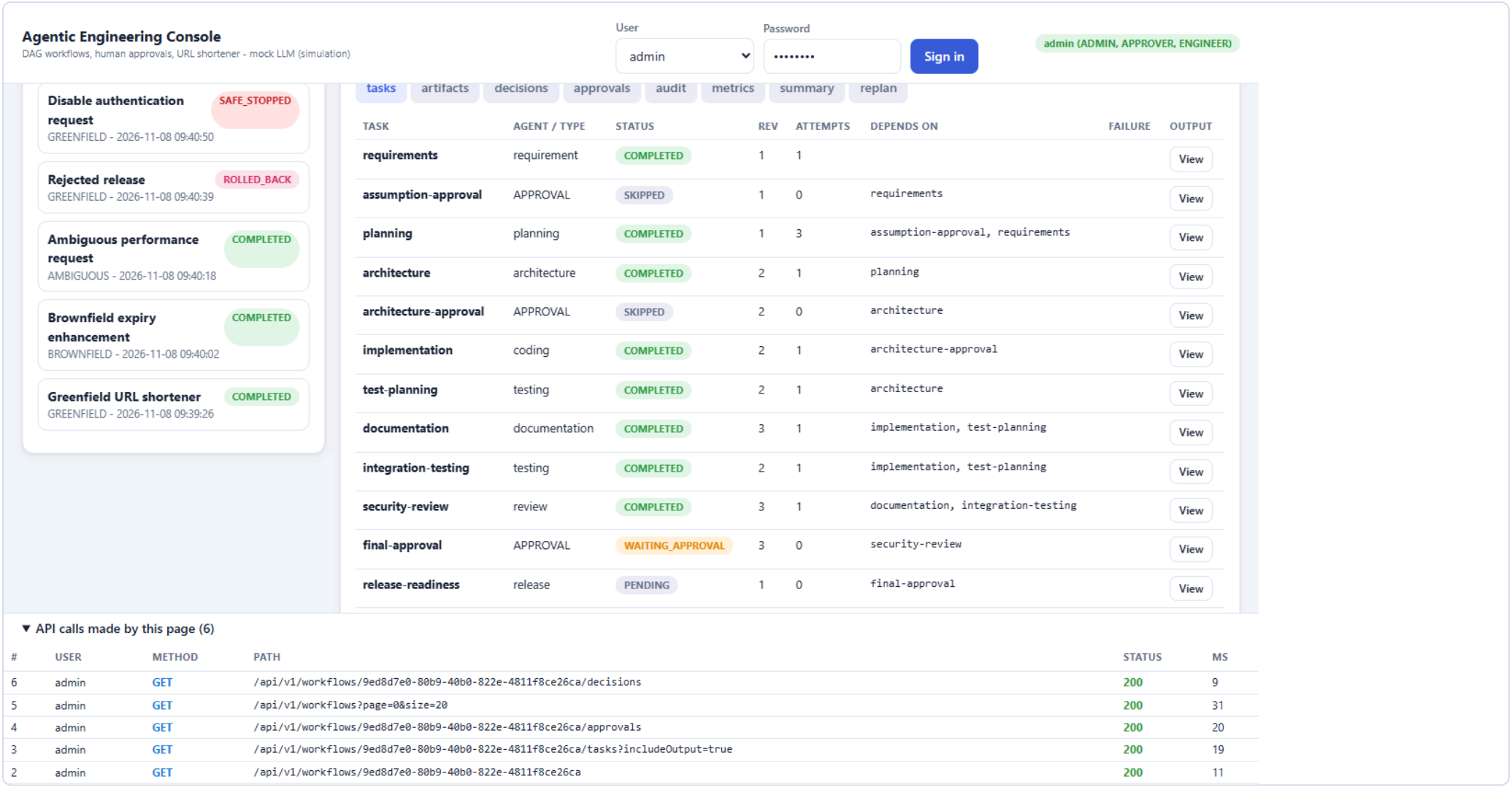
USER	METHOD	API CALL	HTTP	MS
admin	POST	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approve	200	164
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca	200	11

USER	METHOD	API CALL	HTTP	MS
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/tasks?includeOutput=true	200	19
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/approvals	200	20
admin	GET	/api/v1/workflows?page=0&size=20	200	31
admin	GET	/api/v1/workflows/9ed8d7e0-80b9-40b0-822e-4811f8ce26ca/decisions	200	9

DYNAMIC REPLANNING

Step 41. Revisions after replan

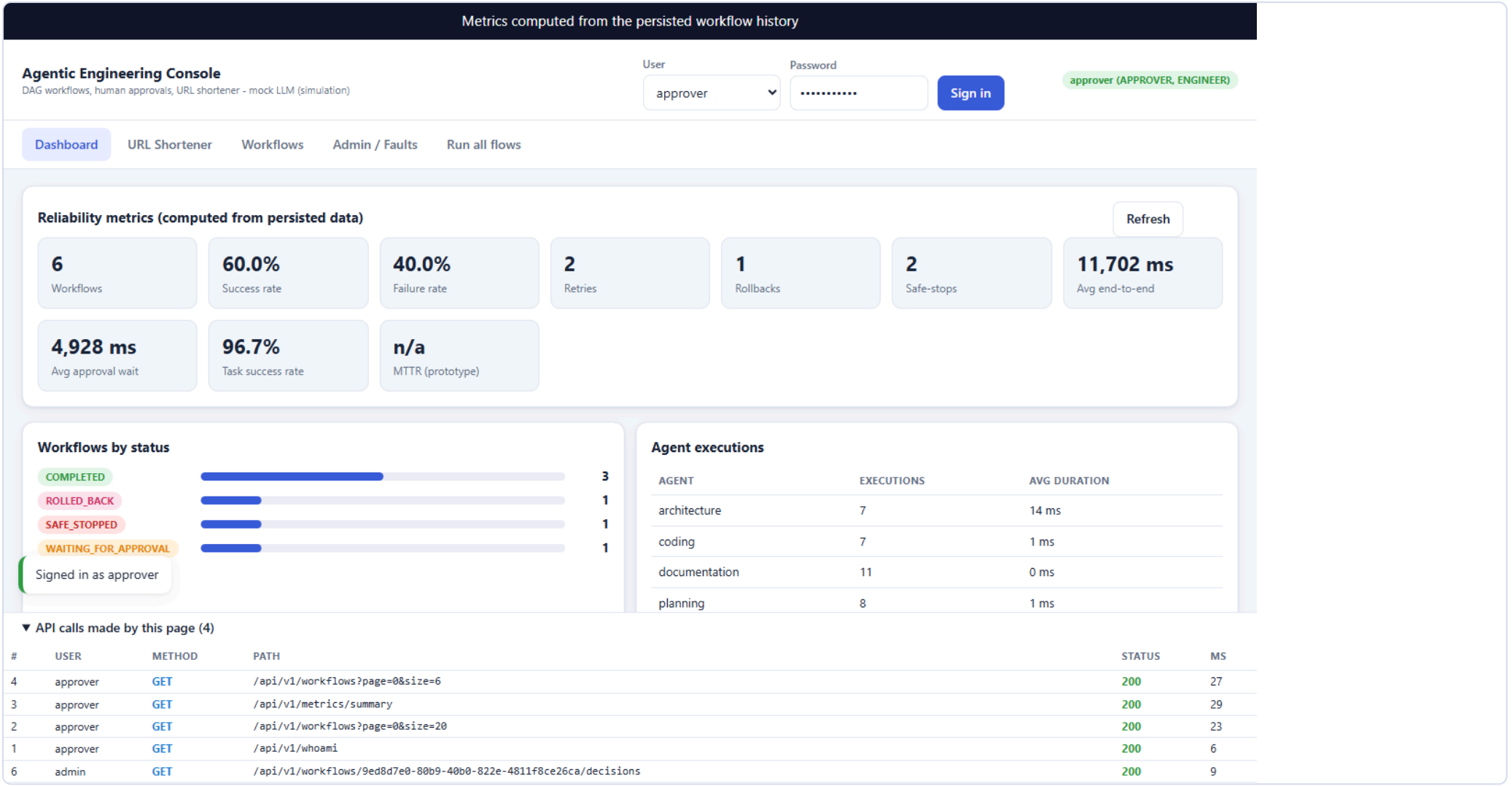
architecture now shows revision 2 while requirements and planning remain revision 1 (preserved).



OBSERVABILITY

Step 42. Metrics dashboard

GET /api/v1/metrics/summary: success and failure rate, retries, rollbacks, safe-stops, end-to-end latency, approval wait, task success rate and MTTR, all derived from stored data.



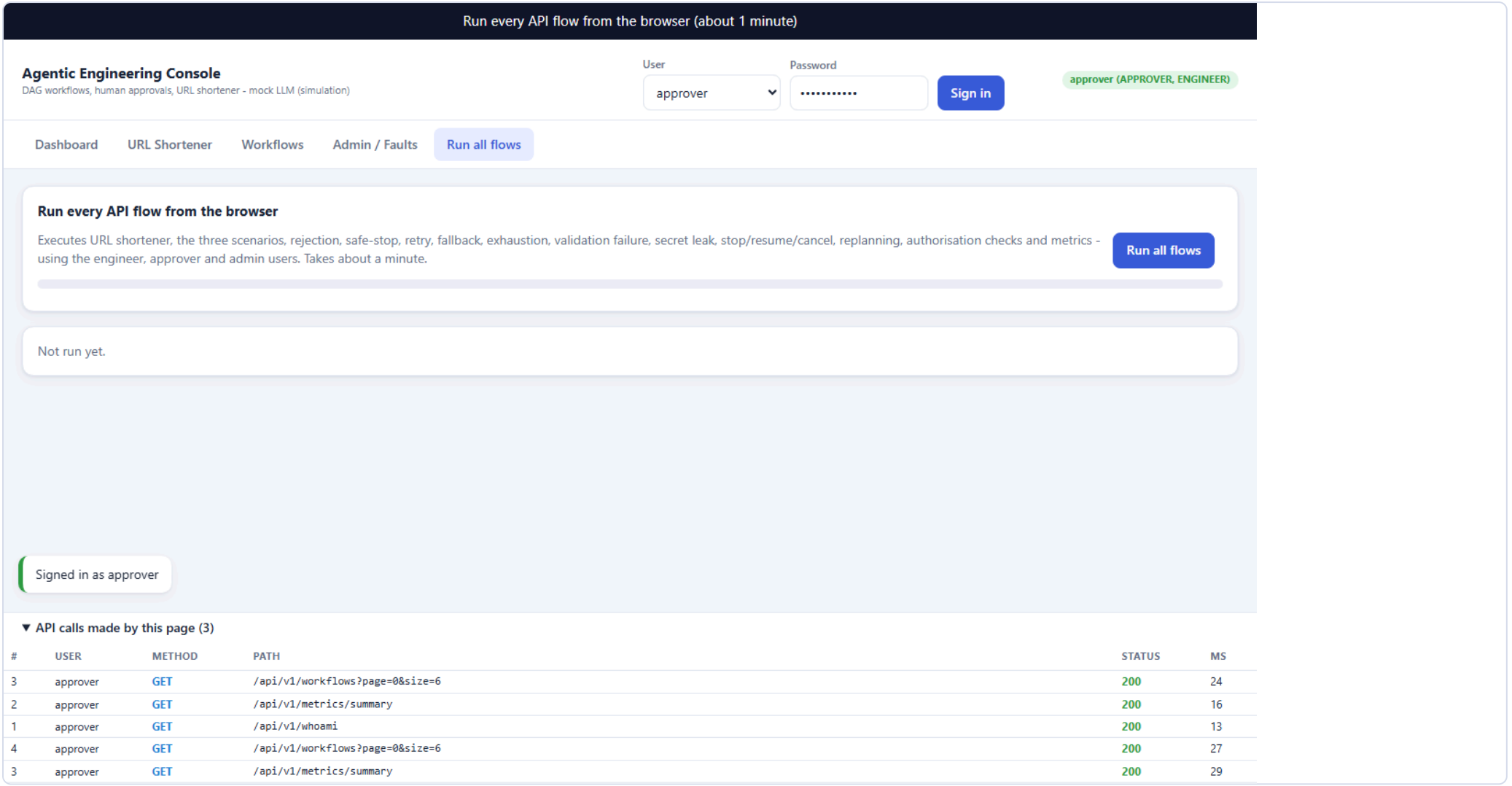
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/whoami	200	6
approver	GET	/api/v1/workflows?page=0&size=20	200	23

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/metrics/summary	200	29
approver	GET	/api/v1/workflows?page=0&size=6	200	27

RUN ALL FLOWS FROM THE BROWSER

Step 43. Run all flows: ready

One button executes every API flow with the engineer, approver and admin users.



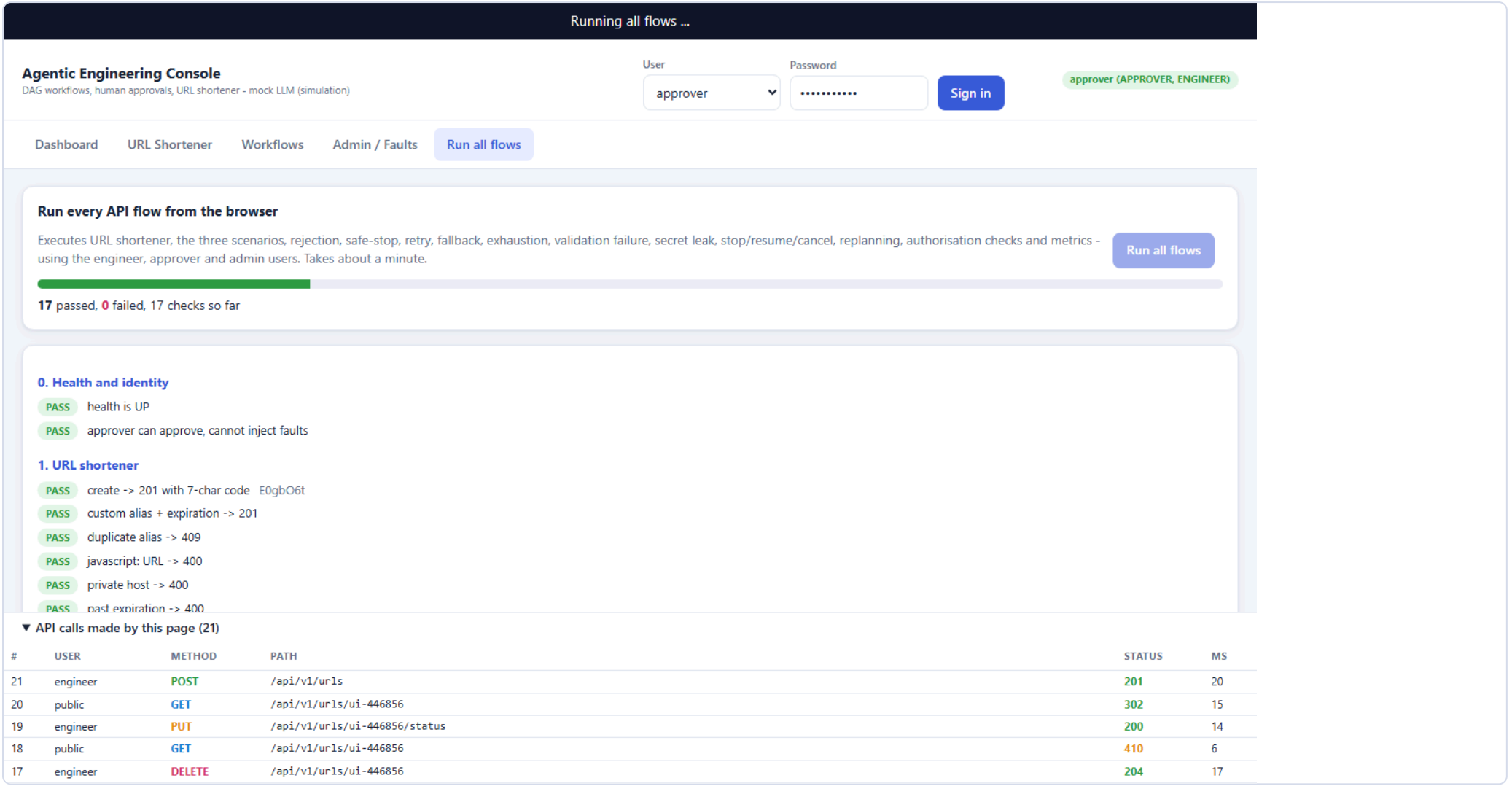
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/whoami	200	13
approver	GET	/api/v1/metrics/summary	200	16

USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/workflows?page=0&size=6	200	24

RUN ALL FLOWS FROM THE BROWSER

Step 44. Run all flows: in progress

Progress bar and live results while the flows execute.



USER	METHOD	API CALL	HTTP	MS
admin	DELETE	/api/v1/admin/fault-injection	200	8
engineer	GET	/api/v1/health	200	5

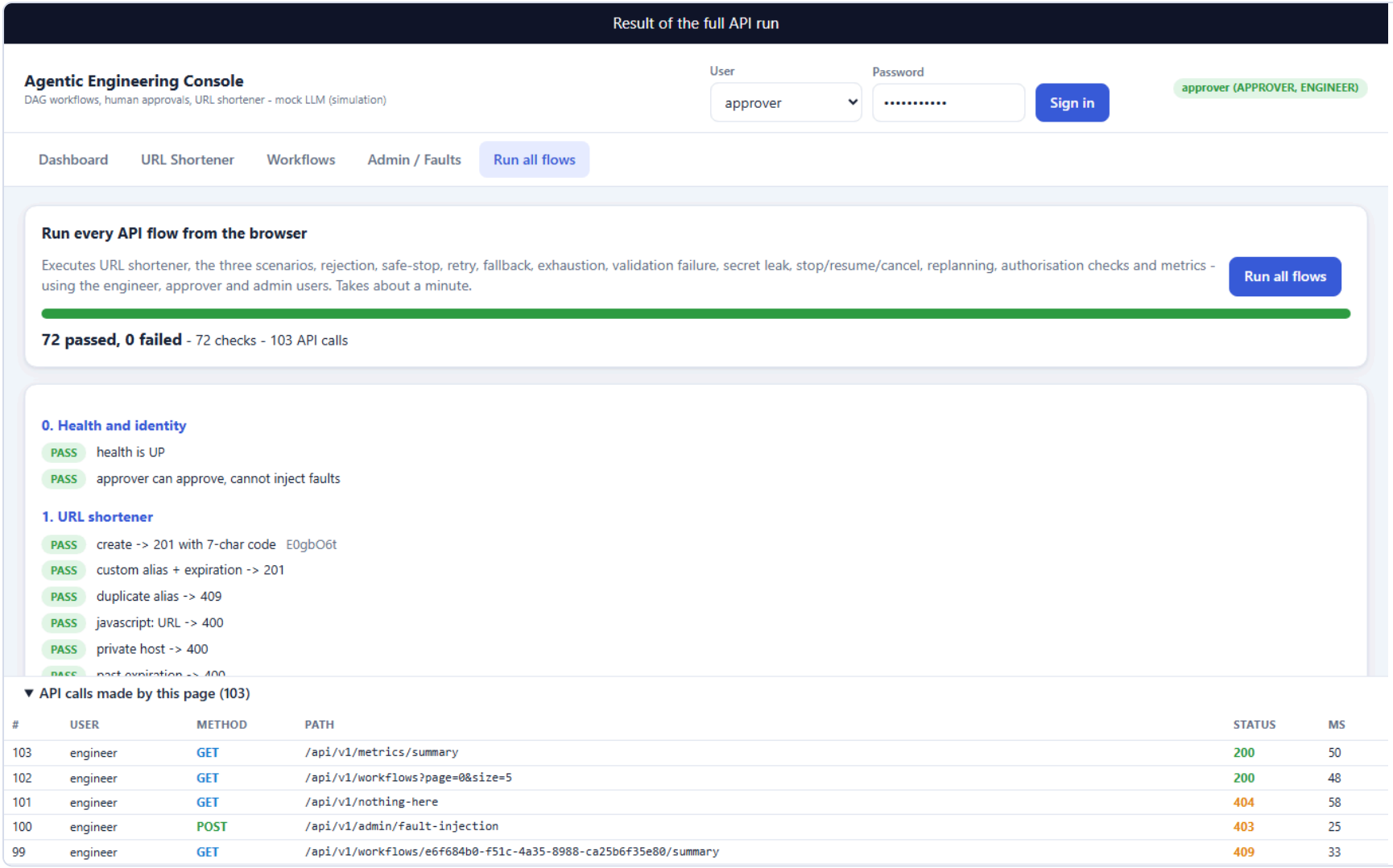
USER	METHOD	API CALL	HTTP	MS
approver	GET	/api/v1/whoami	200	7
engineer	POST	/api/v1/urls	201	11
engineer	POST	/api/v1/urls	201	13
engineer	POST	/api/v1/urls	409	11
engineer	POST	/api/v1/urls	400	11
engineer	POST	/api/v1/urls	400	16
engineer	POST	/api/v1/urls	400	16
anonymous	POST	/api/v1/urls	401	0
public	GET	/api/v1/urls/ui-446856	302	29
public	GET	/api/v1/urls/ui-446856	302	26
engineer	GET	/api/v1/urls/ui-446856/analytics	200	16
engineer	GET	/api/v1/urls?page=0&size=5	200	22
engineer	GET	/api/v1/urls/ui-446856/details	200	17
public	GET	/api/v1/urls/nope-nope-x	404	19

... and 5 more calls in this step

RUN ALL FLOWS FROM THE BROWSER

Step 45. Run all flows: result summary

Every check passed. The page made all of these API calls itself.



USER	METHOD	API CALL	HTTP	MS
public	GET	/api/v1/urls/soon-671438	410	10
engineer	POST	/api/v1/workflows	201	41

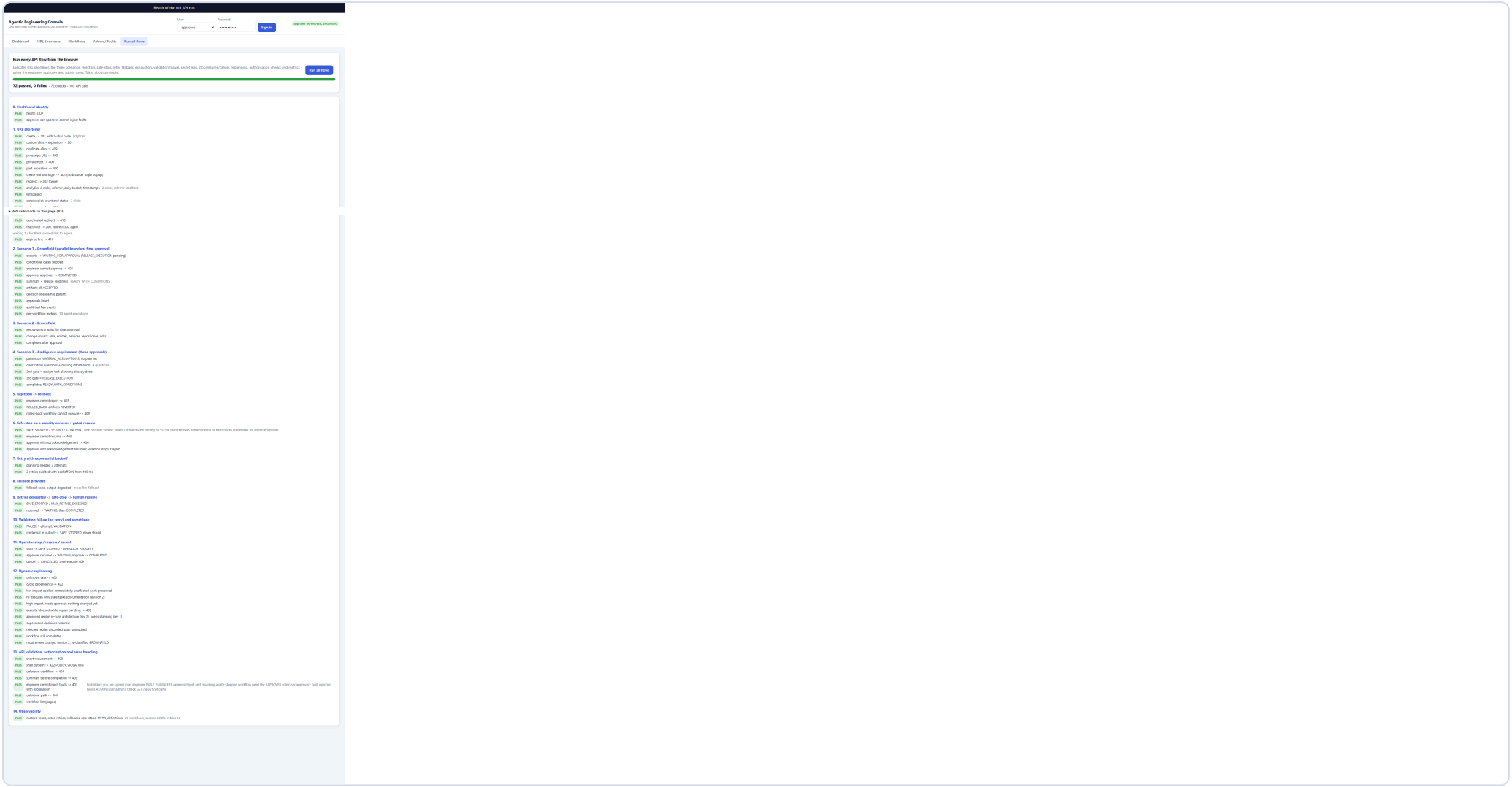
USER	METHOD	API CALL	HTTP	MS
engineer	POST	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/execute	200	228
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/tasks?includeOutput=true	200	10
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/tasks?includeOutput=true	200	12
engineer	POST	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/approve	403	12
approver	POST	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/approve	200	285
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/summary	200	17
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/artifacts	200	10
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/decisions	200	9
engineer	GET	/api/v1/workflows/d2e76920-510c-4c1b-8589-90357af22eb0/approvals	200	15
engineer	GET	/api/v1/audit/workflows/d2e76920-510c-4c1b-8589-90357af22eb0	200	14
engineer	GET	/api/v1/metrics/workflows/d2e76920-510c-4c1b-8589-90357af22eb0	200	20
engineer	POST	/api/v1/workflows	201	136
engineer	POST	/api/v1/workflows/74b3ad5d-8cf0-49be-8d0f-9d281bacbf2d/execute	200	1517
engineer	GET	/api/v1/workflows/74b3ad5d-8cf0-49be-8d0f-9d281bacbf2d/tasks?includeOutput=true	200	25

... and 87 more calls in this step

RUN ALL FLOWS FROM THE BROWSER

Step 46. Run all flows: complete list of checks

Full list of checks (full-page capture).



APPENDIX A

Checks executed by "Run all flows"

72 passed, 0 failed, 72 checks.

0. Health and identity	
PASS	health is UP
PASS	approver can approve, cannot inject faults
1. URL shortener	
PASS	create -> 201 with 7-char code E0gbO6t
PASS	custom alias + expiration -> 201
PASS	duplicate alias -> 409
PASS	javascript: URL -> 400
PASS	private host -> 400
PASS	past expiration -> 400
PASS	create without login -> 401 (no browser login popup)
PASS	redirect -> 302 (twice)
PASS	analytics: 2 clicks, referrer, daily bucket, timestamps 2 clicks, referrer localhost
PASS	list (paged)
PASS	details: click count and status 2 clicks
PASS	unknown code -> 404
PASS	deactivate -> 204
PASS	deactivated redirect -> 410
PASS	reactivate -> 200, redirect 302 again
	waiting 7 s for the 5-second link to expire...
PASS	expired link -> 410
2. Scenario 1 - Greenfield (parallel branches, final approval)	
PASS	execute -> WAITING_FOR_APPROVAL (RELEASE_EXECUTION pending)
PASS	conditional gates skipped
PASS	engineer cannot approve -> 403
PASS	approver approves -> COMPLETED
PASS	summary + release readiness READY_WITH_CONDITIONS
PASS	artifacts all ACCEPTED
PASS	decision lineage has parents
PASS	approvals listed
PASS	audit trail has events

PASS	per-workflow metrics 10 agent executions

3. Scenario 2 - Brownfield

PASS	BROWNFIELD waits for final approval
PASS	change impact: APIs, entities, services, repositories, risks
PASS	completes after approval

4. Scenario 3 - Ambiguous requirement (three approvals)

PASS	pauses on MATERIAL_ASSUMPTIONS, no plan yet
PASS	clarification questions + missing information 4 questions
PASS	2nd gate = design; test planning already done
PASS	3rd gate = RELEASE_EXECUTION
PASS	completes; READY_WITH_CONDITIONS

5. Rejection -> rollback

PASS	engineer cannot reject -> 403
PASS	ROLLED_BACK, artifacts REVERTED
PASS	rolled-back workflow cannot execute -> 409

6. Safe-stop on a security concern + gated resume

PASS	SAFE_STOPPED / SECURITY_CONCERN Task 'security-review' failed: Critical review finding RV-3: The plan removes authentication or hard-codes credentials for admin endpoints
PASS	engineer cannot resume -> 403
PASS	approver without acknowledgement -> 400
PASS	approver with acknowledgement resumes; violation stops it again

7. Retry with exponential backoff

PASS	planning needed 3 attempts
PASS	2 retries audited with backoff 200 then 400 ms

8. Fallback provider

PASS	fallback used, output degraded mock-llm-fallback

9. Retries exhausted -> safe-stop -> human resume

PASS	SAFE_STOPPED / MAX_RETRIES_EXCEEDED
PASS	resumed -> WAITING, then COMPLETED

10. Validation failure (no retry) and secret leak

PASS	FAILED, 1 attempt, VALIDATION
PASS	credential in output -> SAFE_STOPPED, never stored

11. Operator stop / resume / cancel

PASS	stop -> SAFE_STOPPED / OPERATOR_REQUEST

PASS	approver resumes -> WAITING; approve -> COMPLETED
PASS	cancel -> CANCELLED, then execute 409

12. Dynamic replanning

PASS	unknown task -> 400
PASS	cyclic dependency -> 422
PASS	low impact applied immediately; unaffected work preserved
PASS	re-executes only stale tasks (documentation revision 2)
PASS	high impact needs approval; nothing changed yet
PASS	execute blocked while replan pending -> 409
PASS	approved replan re-runs architecture (rev 2), keeps planning (rev 1)
PASS	superseded decisions retained
PASS	rejected replan discarded; plan untouched
PASS	workflow still completes
PASS	requirement change: version 2, re-classified BROWNFIELD

13. API validation, authorisation and error handling

PASS	short requirement -> 400
PASS	shell pattern -> 422 POLICY_VIOLATION
PASS	unknown workflow -> 404
PASS	summary before completion -> 409
PASS	engineer cannot inject faults -> 403 with explanation Forbidden: you are signed in as engineer [ROLE_ENGINEER]. Approve/reject and resuming a safe-stopped workflow need the APPROVER role (user approver); fault injection needs ADMIN (user admin). Check GET /api/v1/whoami.
PASS	unknown path -> 404
PASS	workflow list (paged)

14. Observability

PASS	metrics: totals, rates, retries, rollbacks, safe-stops, MTTR, definitions 20 workflows, success 60.0%, retries 12

APPENDIX B

API coverage (29/29 endpoints)

Endpoints are read from the application's own OpenAPI document and matched against the calls the browser made.

METHOD	ENDPOINT	CALLS	HTTP STATUSES SEEN	COVERED
DELETE	/api/v1/admin/fault-injection	5	200	yes
GET	/api/v1/admin/fault-injection	5	200, 403	yes
POST	/api/v1/admin/fault-injection	8	200, 403	yes
GET	/api/v1/audit/workflows/{workflowId}	4	200	yes
GET	/api/v1/health	1	200	yes
GET	/api/v1/metrics/summary	4	200	yes
GET	/api/v1/metrics/workflows/{workflowId}	2	200	yes
GET	/api/v1/urls	7	200	yes
POST	/api/v1/urls	9	201, 400, 401, 409	yes
DELETE	/api/v1/urls/{shortCode}	2	204	yes
GET	/api/v1/urls/{shortCode}	12	302, 404, 410	yes
GET	/api/v1/urls/{shortCode}/analytics	2	200	yes
GET	/api/v1/urls/{shortCode}/details	1	200	yes
PUT	/api/v1/urls/{shortCode}/status	2	200	yes
GET	/api/v1/whoami	9	200	yes
GET	/api/v1/workflows	50	200	yes
POST	/api/v1/workflows	22	201, 400, 422	yes
GET	/api/v1/workflows/{id}	30	200, 404	yes
GET	/api/v1/workflows/{id}/approvals	30	200	yes
POST	/api/v1/workflows/{id}/approve	20	200, 403	yes
GET	/api/v1/workflows/{id}/artifacts	5	200	yes
POST	/api/v1/workflows/{id}/cancel	1	200	yes
GET	/api/v1/workflows/{id}/decisions	4	200	yes
POST	/api/v1/workflows/{id}/execute	30	200, 400, 403, 409	yes
POST	/api/v1/workflows/{id}/reject	4	200, 403	yes
POST	/api/v1/workflows/{id}/replan	8	200, 400, 422	yes
POST	/api/v1/workflows/{id}/stop	1	200	yes
GET	/api/v1/workflows/{id}/summary	3	200, 409	yes

METHOD	ENDPOINT	CALLS	HTTP STATUSES SEEN	COVERED
GET	/api/v1/workflows/{id}/tasks	46	200	yes